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Lecture 1: Optimal Transport

Lecture 2: Generative Flows

- Experimental Part

Lecture 3: From 1D Processes to Flows

- Experimental Part

Lecture 4: Bayesian Inverse Problems

- Experimental Part

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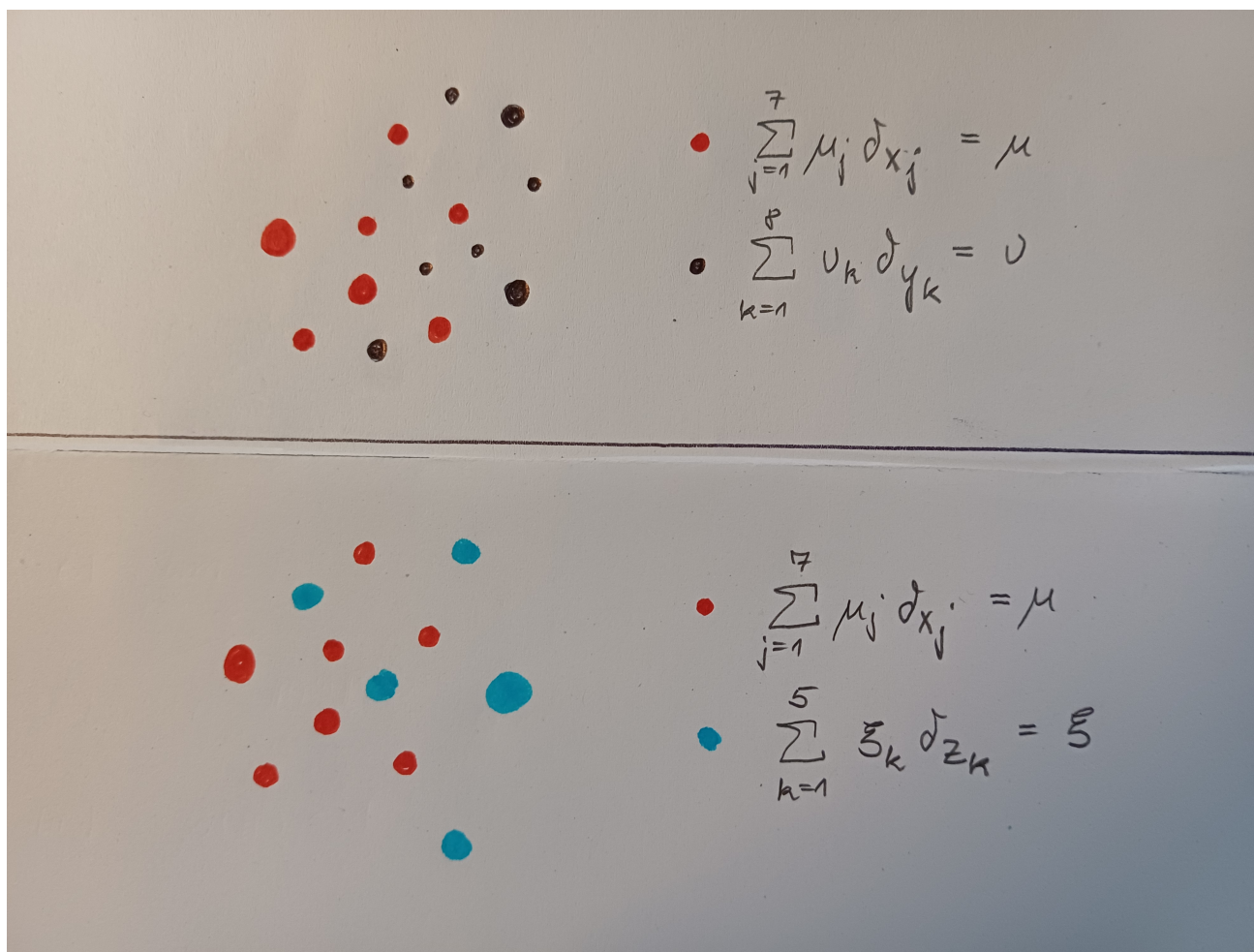
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Question:

- ◆ What is the best way to transport the mass from μ to ν and what costs such a transport?
- ◆ Is ν or ξ nearer to μ ?



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1. Optimal Transport (OT)

1. Spaces of Measures
2. Monge and Kantorovich Problem of OT
3. Discrete OT and its Dual
4. Continuous OT
5. Regularized Optimal Transport

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1. Spaces of Measures

- ◆ $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \mu): \mu : \mathcal{B}(\mathbb{R}^d) \rightarrow \mathbb{R}$ with

$$\mu(\emptyset) = 0,$$

$$\mu\left(\bigcup_k A_k\right) = \sum_{k=1}^{\infty} \mu(A_k)$$

- ◆ $\mathcal{M}(\mathbb{R}^d)$ is a Banach space with **total variation norm**

$$\|\mu\|_{TV} := |\mu|(\mathbb{R}^d) \quad \text{where} \quad |\mu|(A) := \sup_{A=\bigcup_k A_k, A_i \cap A_j = \emptyset} \sum_{k=1}^n |\mu(A_k)|$$

- ◆ pre-dual space of $\mathcal{M}(\mathbb{R}^d)$ is $C_0(\mathbb{R}^d)$, i.e. $C_0(\mathbb{R}^d)' = \mathcal{M}(\mathbb{R}^d)$

$$\|\mu\|_{TV} = \sup_{\|\varphi\|_{\infty} \leq 1} |\langle \varphi, \mu \rangle|, \quad \langle x, \mu \rangle := \int_{\mathbb{R}^d} \varphi(x) \, d\mu(x)$$

where $\|\varphi\|_{\infty} := \sup_{x \in \mathbb{R}^d} |\varphi(x)|$ and

$$\langle x, \mu \rangle := \int_{\mathbb{R}^d} \varphi(x) \, d\mu(x)$$

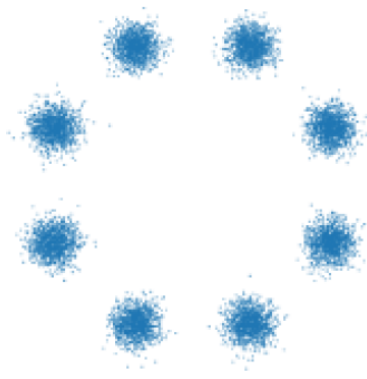
- ◆ **distance** between $\mu, \nu \in \mathcal{M}(\mathbb{R}^d)$:

$$\|\mu - \nu\|_{TV} = \sup_{\|\varphi\|_{\infty} \leq 1} \int_{\mathbb{R}^d} \varphi(x) \, d\mu(x) - \int_{\mathbb{R}^d} \varphi(x) \, d\nu(x)$$

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- ◆ $\mathcal{M}_+(\mathbb{R}^d)$ non-negative measures $\mu : \mathcal{B}(\mathbb{R}^d) \rightarrow \mathbb{R}_{\geq 0}$
- ◆ $\mathcal{P}(\mathbb{R}^d)$ probability measures $\mu : \mathcal{B}(\mathbb{R}^d) \rightarrow \mathbb{R}_{\geq 0}$ with $\mu(\mathbb{R}^d) = 1$

Samples from a probability measures:



$d = 2$



$d = 28^2 = 784$

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Examples:

1. Atomic measures (empirical measures if same weights)

$$\mu = \sum_{i=1}^m \mu_i \delta_{x_i}, \quad \delta_x(A) := \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{otherwise} \end{cases}$$

$$\|\mu\|_{TV} = \mu_1 + \dots + \mu_m = 1$$

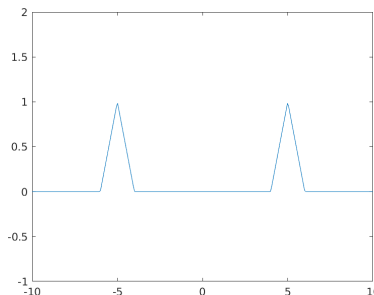
2. Absolutely continuous measures $\mu \in \mathcal{P}(\mathbb{R}^d)$ with density $p \in L_1(\mathbb{R}^d)$

$$\mu(A) = \int_A p(x) \, dx$$

$$\|\mu\|_{TV} = \int_{\mathbb{R}^d} |p(x)| \, dx = 1$$

- ◆ Also, if $\mu = p dx$ and $\nu = q dx$ have disjoint supports 🙄

$$\|\mu - \nu\|_{TV} = \sup_{\|\varphi\|_{\infty} \leq 1} \int_{\mathbb{R}^d} \varphi(x) (p(x) - q(x)) \, dx = 2$$



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2. Monge Problem (1781)



Given $\mu \in \mathcal{P}(\mathbb{R}^d)$, $\nu \in \mathcal{P}(\mathbb{R}^d)$

Find an optimal **transport map** $\hat{T} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that

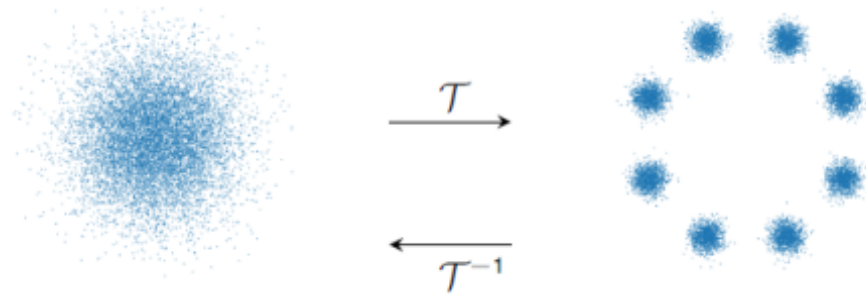
$$\hat{T} \in \operatorname{argmin}_{T \text{ measurable}} \int_{\mathbb{R}^d} c(x, T(x)) \, d\mu(x) \quad \text{subject to} \quad \nu = T_{\#}\mu$$

and then a **good measure** 😊

$$\int_{\mathbb{R}^d} c(x, \hat{T}(x)) \, d\mu(x)$$

with the **push forward measure**

$$T_{\#}\mu := \mu \circ T^{-1}$$



◆ $\int_A f(y) \, d\nu(y) = \int_A f(y) \, d(T_{\#}\mu)(y) = \int_{T^{-1}(A)} f(T(x)) \, d\mu(x)$

◆ in case of density $\mu = p_{\mu} d\lambda$ and a diffeomorphism T :

$$p_{T_{\#}\mu}(y) = p_{\mu}(T^{-1}(y)) |\det \nabla T^{-1}(y)|$$

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Given

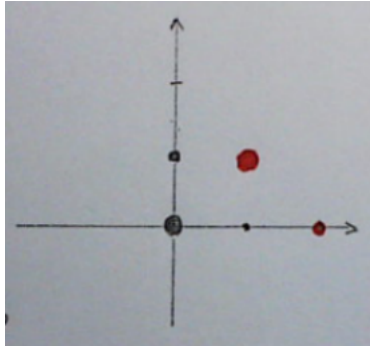
$$\mu = \sum_{i=1}^m \mu_i \delta_{x_i}, \quad \nu = \sum_{j=1}^n \nu_j \delta_{y_j}$$

Find an optimal transport map $\hat{T} : \{x_1, \dots, x_m\} \rightarrow \{y_1, \dots, y_n\}$ such that

$$\hat{T} \in \operatorname{argmin}_T \sum_{i=1}^N c(x_i, T(x_i)) \mu_i \quad \text{s.t.} \quad \nu_j = \sum_{T(x_i)=y_j} \mu_i$$

Example: $x_1 = (0, 0)$, $x_2 = (1, 0)$, $x_3 = (0, 1)$ and $y_1 = (1, 1)$, $y_2 = (2, 0)$

Measures: $\mu = 3\delta_{x_1} + 1\delta_{x_2} + 2\delta_{x_3}$, $\nu = 4\delta_{y_1} + 2\delta_{y_2}$



Costs: $c(x, y) = \|x - y\|^2$

Only possible map: $T(x_1) = y_1$, $T(x_2) = y_1$, $T(x_3) = y_2$

Problem: For the opposite task, i.e. when changing the role of μ and ν there does not exist a transport map T

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More Examples

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Example: For two Gaussians

$$\mu = \mathcal{N}(m_\mu, \Sigma_\mu), \quad \nu = \mathcal{N}(m_\nu, \Sigma_\nu)$$

where $\mathcal{N}(m, \Sigma)$ has the density

$$p(x) := (2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}} e^{-\frac{1}{2}(x-m)^\top \Sigma^{-1}(x-m)},$$

the optimal transport map is given by

$$T(x) = m_\nu + A(x - m_\mu), \quad A := \Sigma_\mu^{-\frac{1}{2}} \left(\Sigma_\mu^{\frac{1}{2}} \Sigma_\nu \Sigma_\mu^{\frac{1}{2}} \right) \Sigma_\mu^{-\frac{1}{2}}$$

Example: For empirical measures

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}, \quad \nu = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}$$

we have just a sorting problem

$$x_{\pi(1)} \leq \dots \leq x_{\pi(n)} \quad y_{\pi(1)} \leq \dots \leq y_{\pi(n)} \quad \text{and} \quad T(x_{\pi(k)}) = y_{\pi(k)}$$

which can be done in $O(n \log n)$

Kantorovich Problem (1942)



Given $\mu \in \mathcal{P}(\mathbb{R}^d), \nu \in \mathcal{P}(\mathbb{R}^d)$

Find an optimal **transport plan** $\hat{\alpha} \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ with given marginals μ and ν

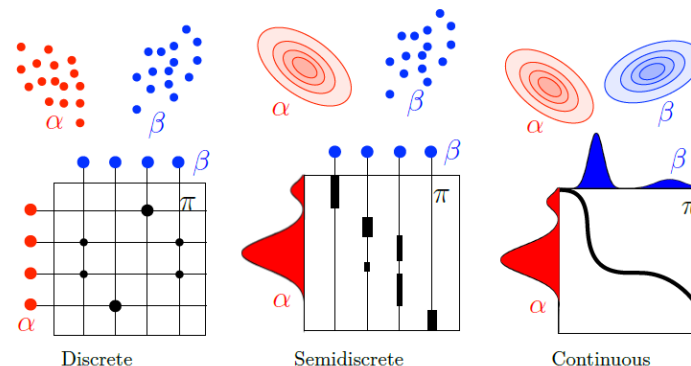
$$\hat{\alpha} \in \operatorname{argmin}_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) d\alpha(x, y)$$

where

- ◆ $\Pi(\mu, \nu) := \{\alpha \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d) : (P_1)_\# \alpha = \mu, (P_2)_\# \alpha = \nu\}$
- ◆ $P_1(x_1, x_2) = x_1, P_2(x_1, x_2) = x_2$
- ◆ **Existence** of a minimizer is ensured if c is lsc and bounded from below

Good measure 😊

$$\operatorname{OT}(\mu, \nu) = \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) d\hat{\alpha}(x, y)$$



: Book Peyré/Cuturi: Computational optimal transport

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Discrete Kantorovich Problem

Given

$$\mu = \sum_{i=1}^m \mu_i \delta_{x_i}, \quad \nu = \sum_{j=1}^n \nu_j \delta_{y_j}$$

Find an optimal **transport plan** $\hat{\alpha} = \sum_{i,j=1}^{m,n} \hat{\alpha}_{ij} \delta_{x_i, y_j}$

Set $\alpha := (\alpha_{ij})_{i,j=1}^{m,n}$ and $c := (c(x_i, y_j))_{i,j=1}^{m,n}$.

OT is a linear optimization problem (solution e.g. by simplex algorithm)

$$\hat{\alpha} \in \operatorname{argmin}_{\alpha_{ij} \geq 0} \sum_{i=1}^m \sum_{j=1}^n c(x_i, y_j) \alpha_{ij}, \quad \hat{\alpha} \in \operatorname{argmin}_{\alpha \geq 0} \langle c, \alpha \rangle$$

subject to

$$\sum_{i=1}^m \alpha_{ij} = \nu_j, \quad j = 1, \dots, n, \quad \alpha^\top 1_m = \nu$$

$$\sum_{j=1}^n \alpha_{ij} = \mu_i, \quad i = 1, \dots, m, \quad \alpha 1_n = \mu$$

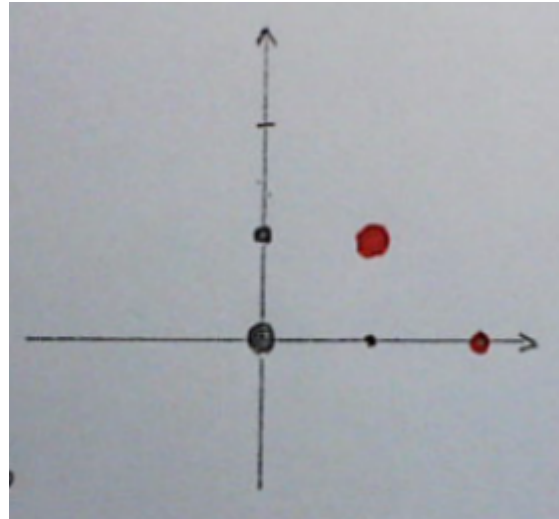
	ν_1	$\dots$	ν_j	$\dots$	ν_m
μ_1	$\alpha_{1,1}$	$\dots$	$\alpha_{1,j}$	$\dots$	$\alpha_{1,m}$
$\vdots$			$\vdots$		$\vdots$
μ_i	$\alpha_{i,1}$	$\dots$	$\alpha_{i,j}$	$\dots$	$\alpha_{i,m}$
$\vdots$			$\vdots$		$\vdots$
μ_n	$\alpha_{n,1}$	$\dots$	$\alpha_{n,j}$	$\dots$	$\alpha_{n,m}$

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Examples

Example 1: $x_1 = (0, 0)$, $x_2 = (1, 0)$, $x_3 = (0, 1)$ and $y_1 = (1, 1)$, $y_2 = (2, 0)$

Measures: $\mu = 3\delta_{x_1} + 1\delta_{x_2} + 2\delta_{x_3}$, $\nu = 4\delta_{y_1} + 2\delta_{y_2}$



Optimal (only possible) map: $T(x_1) = y_1$, $T(x_2) = y_1$, $T(x_3) = y_2$

Optimal plan:

	y_1	y_2
x_1	2	4
x_2	1	1
x_3	1	5

cost $\|x - y\|^2$

	4	2
3	2	1
1	0	1
2	2	0

optimal plan $\hat{\alpha}$
 $\text{OT}(\mu, \nu) = 11$

	4	2
3	3	0
1	1	0
2	0	2

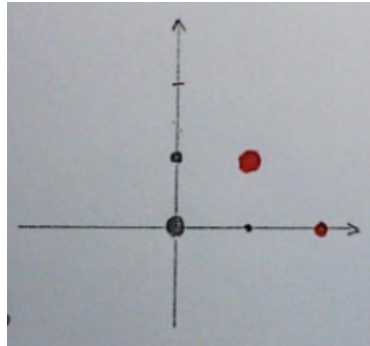
$\alpha = (\text{Id}, T)_\# \mu$
 $\langle c, \alpha \rangle = 17$

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Examples

Example 2: $x_1 = (0, 0)$, $x_2 = (1, 0)$, $x_3 = (0, 1)$ and $y_1 = (1, 1)$, $y_2 = (2, 0)$

Measures: $\mu = 3\delta_{x_1} + 1\delta_{x_2} + 2\delta_{x_3}$, $\nu = 2\delta_{y_1} + 4\delta_{y_2}$



Optimal map: $T(x_1) = y_2$, $T(x_2) = y_2$, $T(x_3) = y_1$

Optimal plan:

	y_1	y_2
x_1	4	2
x_2	1	1
x_3	5	1

costs

	2	4
3	0	3
1	0	1
2	2	0

optimal plan $\hat{\alpha} = (\text{Id}, T)_{\#}\mu$

$$\text{OT}(\mu, \nu) = 17$$

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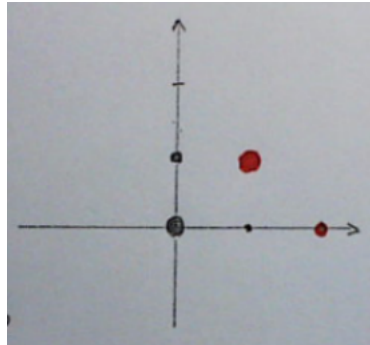
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Examples

Example 3: $x_1 = (1, 1)$, $x_2 = (2, 0)$ and $y_1 = (0, 0)$, $y_2 = (1, 0)$, $y_3 = (0, 1)$ and $y_4 = (1, 1)$, $y_5 = (2, 0)$

Measures: $\mu = 2\delta_{x_1} + 4\delta_{x_2}$, $\nu = 3\delta_{y_1} + 1\delta_{y_2} + 2\delta_{y_3}$



Optimal map: does not exist

Optimal plan:

	y_1	y_2	y_3
x_1	4	1	5
x_2	2	1	1

costs $\|x - y\|^2$

	3	1	2
2	0	0	2
4	3	1	2

optimal plan $\hat{\alpha}$

$\text{OT}(\mu, \nu) = 19$

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For $p \in [1, \infty)$, the space of measures with finite p -th moment is defined by

$$\mathcal{P}_p(\mathbb{R}^d) := \left\{ \mu \in \mathcal{P}(\mathbb{R}^d) : \int_{\mathbb{R}^d} \|x\|^p d\mu(x) < \infty \right\}.$$

For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, the **Wasserstein p -distance** is given by

$$W_p^p(\mu, \nu) := \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^p d\alpha(x, y),$$

which actually defines a **metric**.

The metric space $(\mathcal{P}_p(\mathcal{X}), W_p)$ is called the p -th **Wasserstein space**.

Convergence of $(\mu_n)_n$, $\mu_n \in \mathcal{P}_2(\mathbb{R}^d)$: $W_2(\mu_n, \mu) \rightarrow 0$ as $n \rightarrow \infty$ if and only if we have weak/narrow convergence

$$\mu_n \xrightarrow{*} \mu, i.e. \int \varphi d\mu_n \rightarrow \int \varphi d\mu \quad \forall \varphi \in C_b(\mathbb{R}^d)$$

and

$$\int_{\mathbb{R}^d} \|x\|^2 d\mu_n(x) \rightarrow \int_{\mathbb{R}^d} \|x\|^2 d\mu(x) \quad \text{as } n \rightarrow \infty$$

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Theorem Let $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, where $\mu \ll \lambda$ and $c(x, y) = \|x - y\|^2$. Then

- ◆ Kantorovich problem has a unique solution $\hat{\alpha}$
- ◆ $\hat{\pi} = (I, \hat{T})_{\#}\mu$, where $\hat{T} \in L^2_{\mu}(\mathbb{R}^d, \mathbb{R}^d)$ is the optimal transport map

- ◆ It holds

$$\hat{T}(x) = x - \nabla \varphi(x) = \nabla \psi(x) \quad \text{for } \mu - a.e. \ x,$$

for some lower semi-continuous, convex, differentiable μ -a.e function ψ .

- ◆ Conversely, if ψ is lower semi-continuous, convex and differentiable μ -a.e. with $|\nabla \psi| \in L^2_{\mu}(\mathbb{R}^d, \mathbb{R}^d)$, then

$$T = \nabla \psi$$

is the optimal transport map from μ to $\nu = T_{\#}\mu \in \mathcal{P}_2(\mathbb{R}^d)$.

1. Basic Notation: Spaces of Measures
2. Monge and Kantorovich Problem of OT
3. Dual OT
4. Regularized Optimal Transport

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Discrete OT and its Dual

(P) $\operatorname{argmin}_{\alpha} \langle c, \alpha \rangle$ subject to $\alpha 1_n = \mu, \alpha^\top 1_m = \nu, \alpha \geq 0$ nm variables

(P) $\operatorname{argmin}_{\alpha} \langle c, \alpha \rangle$ subject to $\underbrace{(1_n \otimes I_m)}_{P_1} \alpha = \mu, \underbrace{(I_n \otimes 1_m)}_{P_2} \alpha = \nu, \alpha \geq 0$

$$P_1 = (I_m \quad \dots \quad I_m) \in \mathbb{R}^{m \times nm}$$

$$P_2 = \begin{pmatrix} \underbrace{1 \dots 1}_m & 0 & \dots & 0 \\ 0 & \underbrace{1 \dots 1}_m & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \underbrace{1 \dots 1}_m \end{pmatrix} \in \mathbb{R}^{n \times nm},$$

(P) $\min_{\alpha \geq 0} \max_{\varphi, \psi} \langle c, \alpha \rangle + \langle \mu - P_1 \alpha, \varphi \rangle + \langle \nu - P_2 \alpha, \psi \rangle = \min_{\alpha \geq 0} \max_{\varphi, \psi} L(\alpha, \mu, \nu)$

(D) $\max_{\varphi, \psi} \min_{\alpha \geq 0} \langle c, \alpha \rangle + \langle \mu - P_1 \alpha, \varphi \rangle + \langle \nu - P_2 \alpha, \psi \rangle$

$$= \max_{\varphi, \psi} \min_{\alpha \geq 0} \langle c - P_1^\top \varphi - P_2^\top \psi, \alpha \rangle + \langle \mu, \varphi \rangle + \langle \nu, \psi \rangle =$$

$$= \max_{\varphi_i + \psi_j \leq c_{i,j}} \langle \mu, \varphi \rangle + \langle \nu, \psi \rangle \quad \text{color: red } n + m \text{ variables}$$

If $(c - P_1^\top \varphi - P_2^\top \psi)_{ij} = c_{ij} - \varphi_i - \psi_j < 0$, then first part = $-\infty$

If $(c - P_1^\top \varphi - P_2^\top \psi)_{ij} \geq 0$, then first part = 0, where $\alpha_{ij} = 0$ if $c_{ij} > \varphi_i - \psi_j$

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OT and its Dual (Continuous Setting)

Primal OT:

$$\text{OT}(\mu, \nu) := \min_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} c \, d\alpha$$

Dual OT:

$$\text{OT}(\mu, \nu) = \max_{\substack{(\varphi, \psi) \in C(\mathbb{R}^d)^2 \\ \varphi(x) + \psi(y) \leq c(x, y)}} \int_{\mathbb{R}^d} \varphi \, d\mu + \int_{\mathbb{R}^d} \psi \, d\nu$$

$(\hat{\varphi}, \hat{\psi}) = (\hat{\varphi}, \hat{\varphi}^c)$ with c -transformed function

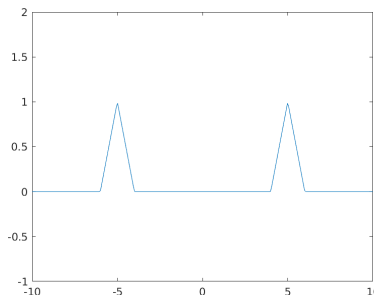
$$\varphi^c(y) = \min_{x \in \mathbb{R}^d} \{c(x, y) - \varphi(x)\}$$

Special case: If $c(x, y) = \|x - y\|$, then $\varphi = -\psi$ is 1-Lipschitz (holds more general $c = d$ for metric space with distance d)

$$W_1(\mu, \nu) = \max_{\varphi \in \text{Lip}_1(\mathbb{R}^d)} \int_{\mathbb{R}^d} \varphi \, d\mu = \max_{\varphi \in \text{Lip}_1(\mathbb{R}^d)} \langle \varphi, \mu \rangle$$

This plays a role in so-called [Wasserstein GANs](#).

Compare to the dual notation of the TV norm: $\|\mu\|_{TV} = \sup_{\|\varphi\|_\infty \leq 1} |\langle \varphi, \mu \rangle|$



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Numerical computation of OT is time consuming for many samples $m = n$:

- ◆ Simplex algorithm $O(n^3 \log n)$
- ◆ Hungarian algorithm $O(n^3)$
- ◆ ...
- ◆ **Next Section: Regularized optimal transport** $O(n^2 \log n)$,
critical for small regularization parameter ε
Cuturi: Sinkhorn algorithm: lightspeed computation of optimal transport
- ◆ Toolboxes:
 - Python Optimal Transport (POT, <https://pythonot.github.io/>):
general toolbox for optimal transport including unbalanced optimal
transport, sliced Wasserstein und Gromov-Wasserstein
 - Geomloss (<https://www.kernel-operations.io/geomloss/>): entropic
optimal transport wit efficient GPU implementation and automatic
differentiation

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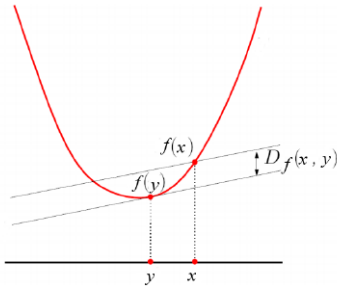
1. Optimal Transport (OT)

1. Spaces of Measures
2. Monge and Kantorovich Problem of OT
3. Discrete OT and its Dual
4. Continuous OT
5. Regularized Optimal Transport

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Bregman distance: $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ convex, lsc and $\text{dom } f \cap (0, +\infty) \neq \emptyset$

$$D_f(x, y) = f(x) - f(y) + \langle \nabla f(y), x - y \rangle$$



Function Name	$f(x)$	$\text{dom } f$	$D_f(x, y)$
Squared Norm	$\frac{1}{2}x^2$	$(-\infty, \infty)$	$\frac{1}{2}(x - y)^2$
Shannon Entropy	$x \log x$	$[0, \infty)$	$x \log \frac{x}{y} - x + y$
Bit Entropy	$x \log x + (1 - x) \log(1 - x)$	$[0, 1]$	$x \log \frac{x}{y} + (1 - x) \log \frac{1-x}{1-y}$
Burg Entropy	$-\log x$	$(0, \infty)$	$\frac{x}{y} - \log \frac{x}{y} - 1$

Examples:

1. $f(x) = \frac{1}{2}\|x\|^2$

$$D_f(x, y) = \frac{1}{2}\|x\|^2 - \frac{1}{2}\|y\|^2 - \langle y, x - y \rangle = \frac{1}{2}\|x - y\|^2$$

2. $f(x) = \langle x, \log x \rangle$ (negative) Shannon entropy = maximal chaos

$$\begin{aligned} D_f(x, y) &= \langle x, \log x \rangle - \langle y, \log y \rangle - \langle \log y + 1, x - y \rangle \\ &= \langle x, \log x - \log y \rangle - \langle x, 1 \rangle + \langle y, 1 \rangle \\ &= \text{KL}(x, y) \end{aligned}$$

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Properties:

- ◆ $D_f(x, y) \geq 0$
- ◆ $D_f(x, y) = 0$ iff $x = y$ in case f is strictly convex.
- ◆ In general not symmetric and does not fulfill a triangular inequality
- ◆ Jointly convex, lsc.
- ◆ If f is strictly convex, D_f is strictly convex in the first argument.
- ◆ If expressions exist

$$\nabla_x D_f(x, y) = \nabla f(x) - \nabla f(y)$$

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$$(P) \quad OT_\varepsilon = \min_{\alpha \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) d\alpha(x, y) + \varepsilon \text{KL}(\alpha, \mu \otimes \nu)$$

◆ Alternatively, we can regularize with the entropy $\text{ent}(\alpha) = \int_{\mathbb{R}^d \times \mathbb{R}^d} \alpha \log(\alpha) dx(x, y)$

Discrete setting:

$$\begin{aligned} (P) \quad OT_\varepsilon(\mu, \nu) &= \min_{\alpha \in \Pi(\mu, \nu)} \langle c, \alpha \rangle + \varepsilon \left(\sum_{i,j} \alpha_{i,j} \log \alpha_{i,j} - \alpha_{i,j} \log(\mu_i \nu_j) - \alpha_{i,j} \right) \\ &= \varepsilon \min_{\alpha \in \Pi(\mu, \nu)} \sum_{i,j} \alpha_{i,j} \log \alpha_{i,j} - \alpha_{i,j} \log(\mu_i \nu_j e^{-c/\varepsilon}) - \alpha_{i,j} \\ &= \varepsilon \min_{\alpha \in \Pi(\mu, \nu)} \text{KL}(\alpha, \underbrace{\text{diag}(\mu) e^{-c/\varepsilon} \text{diag}(\nu)}_K) \end{aligned}$$

Dual form:

$$(P) \quad \min_{\alpha} \max_{\varphi, \psi} \text{KL}(\alpha, K) + \langle P_1 \alpha - \mu, \varphi \rangle + \langle P_2 \alpha - \nu, \psi \rangle$$

$$\begin{aligned} (D) \quad & \max_{\varphi, \psi} \min_{\alpha} \left\{ \text{KL}(\alpha, K) + \langle \alpha, P_1^\top \varphi + P_2^\top \psi \rangle \right\} - \langle \mu, \varphi \rangle - \langle \nu, \psi \rangle \\ &= - \max_{\varphi, \psi} \underbrace{\max_{\alpha} \left\{ \langle \alpha, -P_1^\top \varphi - P_2^\top \psi \rangle - \text{KL}(\alpha, K) \right\}}_{\text{KL}^*(-P_1^\top \varphi - P_2^\top \psi, K)} - \langle \mu, \varphi \rangle - \langle \nu, \psi \rangle \\ &= \min_{\varphi, \psi} \langle K, e^{-P_1^\top \varphi - P_2^\top \psi} \rangle + \langle \mu, \varphi \rangle + \langle \nu, \psi \rangle \end{aligned}$$

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Fenchel conjugate of convex function f :

$$f^*(p) = \max_{x \in \mathbb{R}^d} \{ \langle x, p \rangle - f(x) \}$$

Special case:

$$\text{KL}^*(p, y) = \max_{x \in \mathbb{R}} \{ xp - x \log x + x \log y + x \}$$

Setting the gradient to zero gives

$$0 = p - 1 - \log x + \log y + 1$$

$$\log x = p + \log y$$

$$x = ye^p$$

$$\text{KL}^*(p, y) = ye^p p - ye^p \log y - ye^p p + ye^p \log y + ye^p = ye^p$$

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Sinkhorn Algorithm (Primal)

$$(P) \quad \text{OT}_\varepsilon(\mu, \nu) = \varepsilon \min_{\alpha \in \Pi(\mu, \nu)} \text{KL}(\alpha, K)$$

Sinkhorn algorithm wrt α :

$$\alpha^{(0)} := K,$$

for $r = 0, 1, \dots$

$$\alpha^{(r+\frac{1}{2})} := \text{diag} \left(\frac{\mu}{\alpha^{(r)} \mathbf{1}} \right) \alpha^{(2r)}$$

$$\alpha^{(r+1)} := \alpha^{(r+\frac{1}{2})} \text{diag} \left(\frac{\nu}{(\alpha^{(r+\frac{1}{2})})^\top \mathbf{1}} \right)$$

	ν_1	$\dots$	ν_j	$\dots$	ν_m
μ_1	$\alpha_{1,1}$	$\dots$	$\alpha_{1,j}$	$\dots$	$\alpha_{1,m}$
$\vdots$			$\vdots$		$\vdots$
μ_i	$\alpha_{i,1}$	$\dots$	$\alpha_{i,j}$	$\dots$	$\alpha_{i,m}$
$\vdots$			$\vdots$		$\vdots$
μ_n	$\alpha_{n,1}$	$\dots$	$\alpha_{n,j}$	$\dots$	$\alpha_{n,m}$

◆ See also block-iterative SMART, mirror descent algorithm

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Sinkhorn Algorithm (Dual)

$$(D) \quad \max_{\varphi, \psi} F(\varphi, \psi) = \max_{\varphi, \psi} \langle K, e^{-P_1^\top \varphi - P_2^\top \psi} \rangle + \langle \mu, \varphi \rangle + \langle \nu, \psi \rangle$$

$$P_1 = (I_m \quad \dots \quad I_m) \in \mathbb{R}^{m \times nm}$$

$$P_2 = \begin{pmatrix} \underbrace{1 \dots 1}_m & 0 & \dots & 0 \\ 0 & \underbrace{1 \dots 1}_m & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \underbrace{1 \dots 1}_m \end{pmatrix} \in \mathbb{R}^{n \times nm},$$

Idea:

$$\nabla_{\varphi} F = -\text{diag}(e^{-\varphi}) K \text{diag}(e^{-\psi}) \mathbf{1} + \mu = 0$$

$$\nabla_{\psi} F = -(\text{diag}(e^{-\varphi}) K \text{diag}(e^{-\psi}))^\top \mathbf{1} + \nu = 0$$

Sinkhorn algorithm with $u := e^{-\varphi}$, $v := e^{-\psi}$

$$\nabla_{\varphi} F = 0 : \text{diag}(u) \underbrace{K \text{diag}(v^{(r-1)}) \mathbf{1}} = \mu \quad \rightarrow \quad u^{(r)} = \frac{\mu}{K v^{(r-1)}}$$

$$\nabla_{\psi} F = 0 : \text{diag}(v) \underbrace{K^\top \text{diag}(u^{(r)}) \mathbf{1}} = \nu \quad \rightarrow \quad v^{(r)} = \frac{\nu}{K^\top u^{(r)}}$$

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Sinkhorn Algorithm (Dual)

Relation to α : set gradient of Lagrangian

$$L(\alpha, \varphi, \psi) := \text{KL}(\alpha, K) + \langle P_1 \alpha - \mu, \varphi \rangle + \langle P_2 \alpha - \nu, \psi \rangle$$

wrt α to 0

$$0 = \log \alpha - \log K + P_1^T \varphi + P_2^T \psi \iff \log \alpha = \log K - P_1^T \varphi - P_2^T \psi$$

$$\alpha = \text{diag}(u) K \text{diag}(v)$$

Remark on SMART (simultaneous multiplicative algebraic reconstruction technique) and MIRROR DESCENT algorithm:

GENERAL TASK: $A^T \mathbf{1} = \mathbf{1}$, $A \geq 0$ stochastic matrix, $\mathbf{b} \geq 0$

(1) $\min_{\mathbf{x}} \text{KL}(\mathbf{x} \parallel \mathbf{y})$ st. $A\mathbf{x} = \mathbf{b}$
 $\mathcal{C} = \{\mathbf{x} : A\mathbf{x} = \mathbf{b}\}$ KL projection onto affine set

(2) $\min_{\mathbf{x} \in \mathcal{C}} \text{KL}(\mathbf{x} \parallel \mathbf{b})$
 derivation of alg: $\nabla \text{KL}(A\mathbf{x}, \mathbf{b}) = A^T \log \frac{A\mathbf{x}}{\mathbf{b}}$
 $0 = A^T \log \frac{A\mathbf{x}}{\mathbf{b}}$
 $\mathbf{1} = e^{-A^T \log \frac{A\mathbf{x}}{\mathbf{b}}}$
 $\mathbf{x} = \mathbf{x} \circ e^{-A^T \log \frac{A\mathbf{x}}{\mathbf{b}}}$ fixed point eq.
 $\mathbf{x}^{(n+1)} = \mathbf{x}^{(n)} \circ e^{-A^T \log \frac{A\mathbf{x}^{(n)}}{\mathbf{b}}}$ Picard Q.
 SMART (Sinkhorn)

(3) $\min_{\mathbf{x} \in \mathcal{C}} f(\mathbf{x})$
 $f(\mathbf{x}) = \text{KL}(\mathbf{x}, \mathbf{x}^{(1)}) + \langle \nabla f(\mathbf{x}^{(1)}), \mathbf{x} - \mathbf{x}^{(1)} \rangle$
 $\mathbf{x}^{(n+1)} = \arg \min_{\mathbf{x} \in \mathcal{C}} \text{KL}(\mathbf{x}, \mathbf{x}^{(n)}) + \langle \nabla f(\mathbf{x}^{(n)}), \mathbf{x} - \mathbf{x}^{(n)} \rangle$
 $\mathbf{x}^{(n+1)} = \arg \min_{\mathbf{x} \in \mathcal{C}} \text{KL}(\mathbf{x}, \mathbf{x}^{(n)}) + \langle \nabla f(\mathbf{x}^{(n)}), \mathbf{x} - \mathbf{x}^{(n)} \rangle$
 Mirror descent alg (MDA) (Mellouli/Vucelja ASPS, Beck/Teboulle 2003)
 $\mathbf{x}^{(n+1)} = \arg \min_{\mathbf{x} \in \mathcal{C}} \text{KL}(\mathbf{x}, \mathbf{x}^{(n)}) + \langle \nabla f(\mathbf{x}^{(n)}), \mathbf{x} - \mathbf{x}^{(n)} \rangle$
 $\mathbf{x}^{(n+1)} = \arg \min_{\mathbf{x} \in \mathcal{C}} \text{KL}(\mathbf{x}, \mathbf{x}^{(n)}) + \langle \nabla f(\mathbf{x}^{(n)}), \mathbf{x} - \mathbf{x}^{(n)} \rangle$
 Convergence: - general MDA $O(\frac{1}{\sqrt{n}})$
 - accelerated DPB: $O(\frac{1}{\sqrt{n}})$ for special f (homotopy scaling prop. of DPB)

SMART for OT_E : $A = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix}$ rescale by $\frac{1}{\sqrt{2}}$ to get $A^T \mathbf{1} = \mathbf{1}$

$\mathbf{T}^{(n+1)} = \text{diag}\left(\frac{\mathbf{1}}{(\mathbf{T}^{(n)})^T \mathbf{1}}\right) \mathbf{T}^{(n)} \text{diag}\left(\frac{\mathbf{1}}{(\mathbf{T}^{(n)})^T \mathbf{1}}\right)^{1/2}$

- this alg is 1/2 slower than Sinkhorn alg.
 - it works also with $\kappa < 1$ (not $\kappa = 1$) instead of $\kappa = \frac{1}{2}$;
 then it is 1/2 slower than Sinkhorn alg. Question 1: Why?

GENERALIZATION
 $\mathbf{C} = \mathbf{C}_1 \cap \mathbf{C}_2 \cap \dots \cap \mathbf{C}_M$, $\mathbf{C}_k = \{\mathbf{x} : A_k \mathbf{x} = \mathbf{b}_k\}$, $A_k, \mathbf{b}_k \geq 0$, $A_k^T \mathbf{1} = \mathbf{1}$
 (4) $\arg \min_{\mathbf{x} \in \mathbf{C}} \text{KL}(\mathbf{x}, \mathbf{y})$ see (1) and (2)
 Iterative information projection: $\mathbf{x}^{(0)} = \mathbf{y}$
 $\mathbf{x}^{(n+1)} = \mathbf{P}_{\mathbf{C}_n}^{\text{KL}}(\mathbf{x}^{(n)})$ $\mathbf{C}_{2n+1} = \mathbf{C}_2$ (periodic projection)
 - convergence for affine sets; not for general convex sets (Bregman alg. for convex sets...)
 Generalized iterative scaling (GIS): $\mathbf{x}^{(0)} = \mathbf{y}$
 $\mathbf{x}^{(n+1)} = \mathbf{x}^{(n)} \circ e^{-A^T \log \frac{A \mathbf{x}^{(n)}}{\mathbf{b}}}$ one SMART step
 BI-SMART: $\mathbf{x}^{(n+1)} = \mathbf{x}^{(n)} \circ e^{-A^T \log \frac{A \mathbf{x}^{(n)}}{\mathbf{b}}}$

BI-SMART for OT_E : $\mathbf{T}^{(n+1)} = \mathbf{T}^{(n)} \circ \text{diag} \frac{\mathbf{1}}{(\mathbf{T}^{(n)})^T \mathbf{1}}$
 - Sinkhorn alg.
 $\mathbf{T}^{(n+2)} = \text{diag} \frac{\mathbf{1}}{(\mathbf{T}^{(n+1)})^T \mathbf{1}} \mathbf{T}^{(n+1)}$

Question 2: BI-SMART for unbalanced OT: $(1-\epsilon) \text{KL}(\mathbf{x}, \mathbf{y}) + \epsilon \text{KL}(A\mathbf{x}, \mathbf{b})$
 does not converge in the obvious form
 Alternatives?
 BI-SMART in case $\mathbf{C} = \emptyset$ (old problem, eq. of Byrne)

Convergence of Sinkhorn Algorithm

On $\mathbb{R}_{>0}^d$ consider the equivalence relation

$$x \sim x' \iff \exists t > 0 : x = t x'.$$

The quotient space, called **positive projective space**

$$\mathbb{P}_{>0}^d := \mathbb{R}_{>0}^d / \sim$$

can be endowed with a vector space structure using the following operations:

- ◆ Addition (internal operation): $x + y := x \odot y$,
- ◆ Scalar multiplication (external operation): $\gamma \cdot x := x^\gamma$,

with component-wise product $\odot$. Using the **Hilbert projective metric**:

$$d_H(x, y) := \log \left(\frac{\max_i x_i / y_i}{\min_i x_i / y_i} \right), \quad x, y \in \mathbb{P}_{>0}^d.$$

the space $\mathbb{P}_{>0}^d$ becomes a complete metric space.

Linear convergence of Sinkhorn algorithm:

$$d_H(u^{(r+1)}, u^*) \leq \lambda(K)^2 d_H(u^{(r)}, u^*)$$

with

$$\lambda(K) := \sup \left\{ \frac{d_H(Kx, Ky)}{d_H(x, y)} : x \not\sim y \right\}$$

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Further Topics

- ◆ Unbalanced optimal transport
- ◆ Sliced Wasserstein distances
- ◆ Gromov-Wasserstein distances
- ◆ Partial Optimal Transport
- ◆ ...

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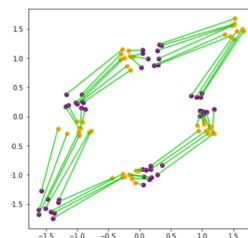
Given:

- ◆ positive measures $\mu, \nu \in \mathcal{M}_+(\mathbb{R}^d)$.
- ◆ cost function $c : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}_+$.
- ◆ divergence measure $D_f(\cdot|\cdot)$ (e.g., KL-divergence, Total Variation).
- ◆ regularization parameters $\tau_1, \tau_2 > 0$.

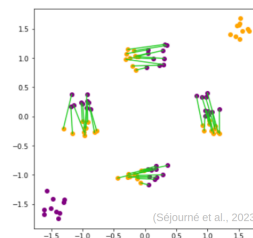
Find **optimal coupling** α :

$$\begin{aligned} \text{UOT}(\mu, \nu) = \min_{\alpha \in \mathcal{M}_+(\mathbb{R}^d \times \mathbb{R}^d)} & \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) d\alpha(x, y) \\ & + \tau_1 D_f((P_1)_\# \alpha | \mu) + \tau_2 D_f((P_2)_\# \alpha | \nu) \end{aligned}$$

- ◆ Compares measures with different mass.
- ◆ Robust to outliers.



(a) OT matching



(b) Unbalanced OT matching

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Given: $\mu, \nu \in \mathcal{P}_p(\mathbb{R})$

Define 1d Projection: $P_\theta(x) = \theta^\top x$ for $\theta \in \mathbb{S}^{d-1}$ (unit sphere)
 $\Rightarrow P_{\theta, \#}\mu, P_{\theta, \#}\nu \in \mathcal{P}_p(\mathbb{R})$

Introduce:

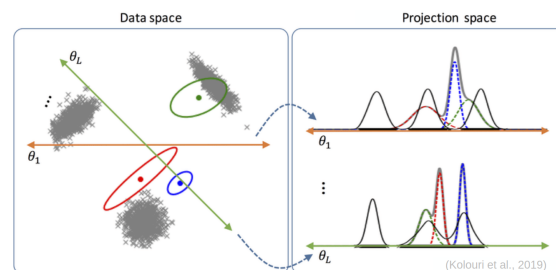
$$SW_p^p(\mu, \nu) = \int_{\mathbb{S}^{d-1}} W_p^p(P_{\theta, \#}\mu, P_{\theta, \#}\nu) d\xi(\theta),$$

where $\xi \sim \mathcal{U}(\mathbb{S}^{d-1})$ is the uniform measure on the unit sphere $\mathbb{S}^{d-1}$.

Calculate Monte Carlo Estimate:

$$SW_p^p(\mu, \nu) \approx \sum_{l=1}^L W_p^p(P_{\theta_l, \#}\mu, P_{\theta_l, \#}\nu), \quad \theta_l \sim \mathcal{U}(\mathbb{S}^{d-1}).$$

- ◆ Fast 1d Wasserstein Computation via Sorting ($\mathcal{O}(n \log n)$)
- ◆ Fast SW approximation ($\mathcal{O}(L n \log n)$)



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Given metric measure spaces:

$$\mathbb{X} = (X, d_X, \mu), \quad (X, d_X) \text{ compact metric space, } \mu \in \mathcal{P}(X),$$

$$\mathbb{Y} = (Y, d_Y, \nu), \quad (Y, d_Y) \text{ compact metric space, } \nu \in \mathcal{P}(Y).$$

Find optimal **GW transport plan** $\hat{\alpha} \in \mathcal{P}(X \times Y)$:

$$\hat{\alpha} \in \operatorname{argmin}_{\alpha \in \Pi(\mu, \nu)} \iint_{(X \times Y)^2} |d_X(x, x') - d_Y(y, y')|^2 d\alpha(x, y) d\alpha(x', y').$$

$$\operatorname{GW}^2(\mathbb{X}, \mathbb{Y}) = \iint_{(X \times Y)^2} |d_X(x, x') - d_Y(y, y')|^2 d\hat{\alpha}(x, y) d\hat{\alpha}(x', y').$$



→ GW vanishes only for isomorphic spaces and allows for interpolating geometries.

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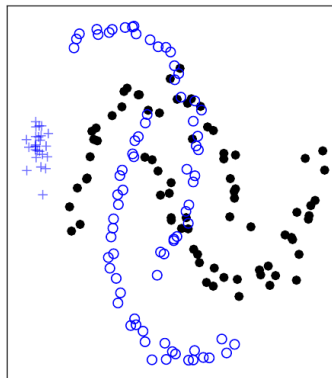
For positive μ, ν , and $s \leq \min\{\|\mu\|_{\text{TV}}, \|\nu\|_{\text{TV}}\}$, consider the **partial plans**

$$\Pi_{\leq}(\mu, \nu) := \left\{ \alpha \in \mathcal{M}_+(X \times Y) : (P_1)_{\#} \alpha \leq \mu, (P_2)_{\#} \alpha \leq \nu, \|\alpha\|_{\text{TV}} = s \right\}.$$

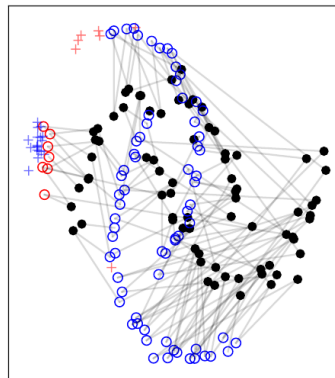
Partial (Gromov–)Wasserstein Distance:

$$\text{PW}_p^p(\mu, \nu) := \inf_{\alpha \in \Pi_{\leq}(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^p d\alpha(x, y)$$

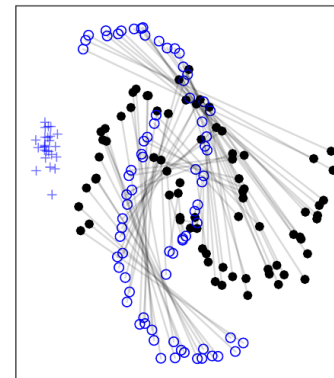
$$\text{PGW}_p^p(\mathbb{X}, \mathbb{Y}) := \inf_{\alpha \in \Pi_{\leq}(\mu, \nu)} \iint_{(X \times Y)^2} |d_X(x, x') - d_Y(y, y')|^p d\alpha(x, y) d\alpha(x', y')$$



source/target



Wasserstein



Gromov–Wasserstein

(Chapel et al., 2020)

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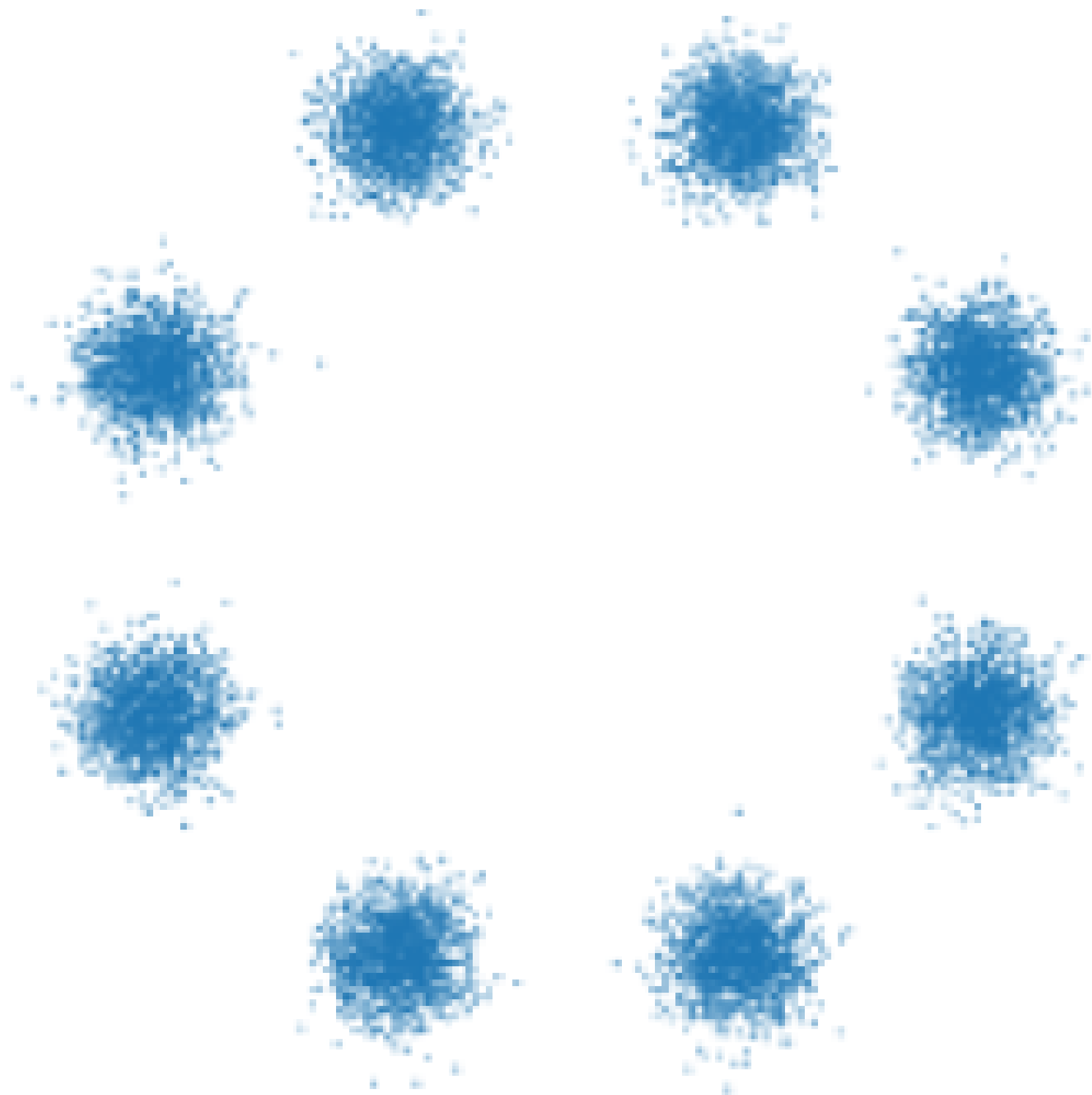
• $\sum_{j=1}^7 \mu_j \delta_{x_j} = \mu$

• $\sum_{k=1}^8 \nu_k \delta_{y_k} = \nu$

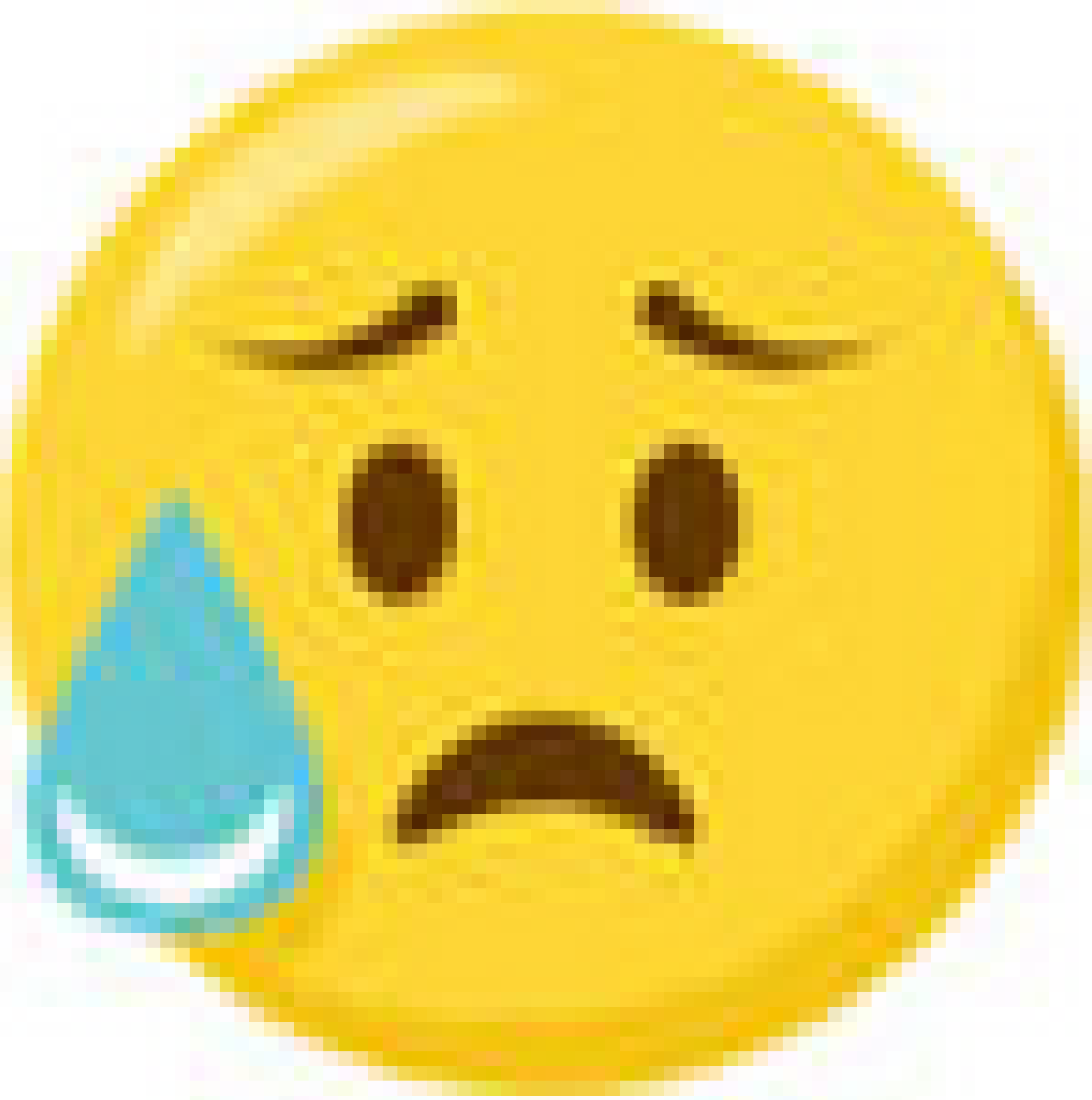


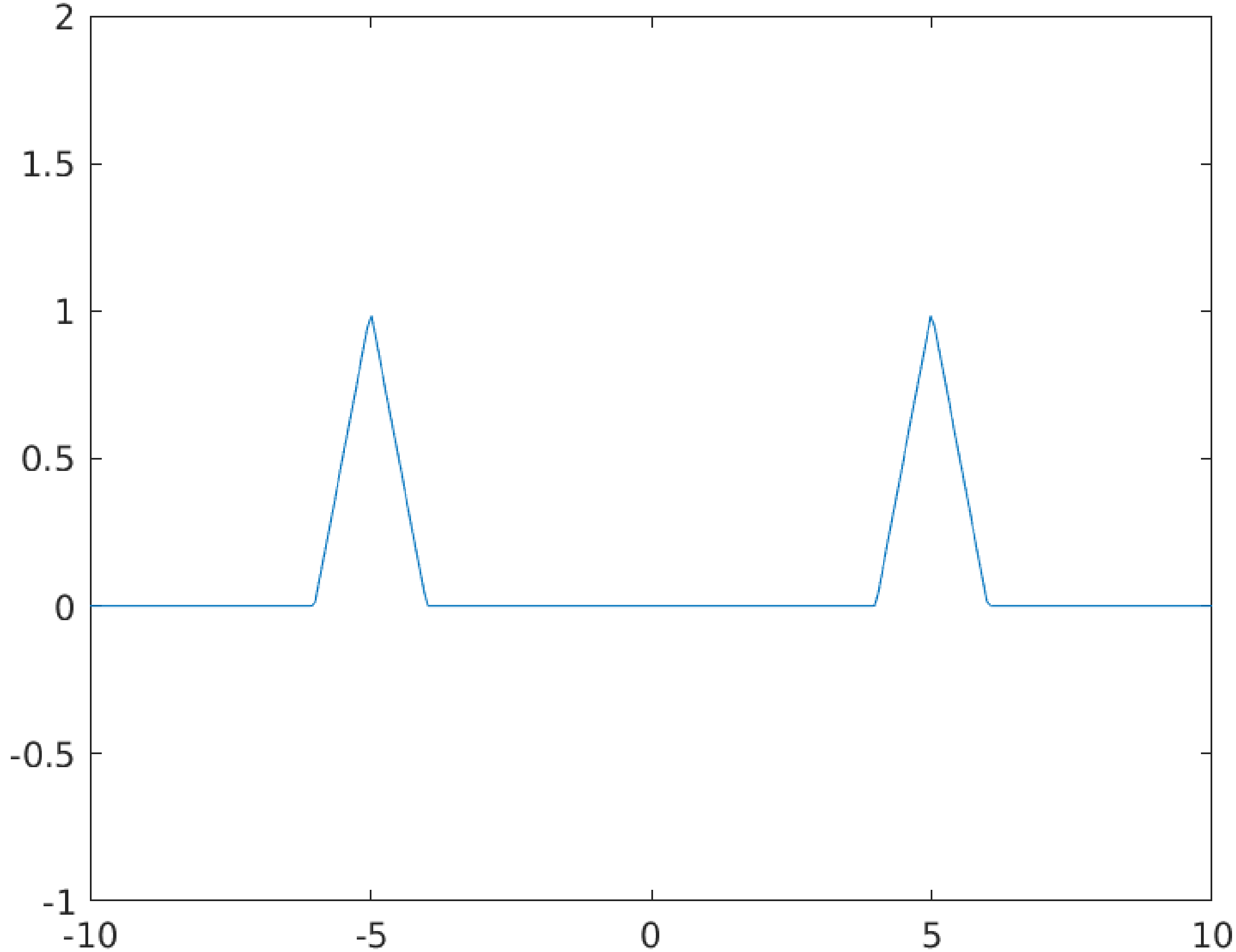
• $\sum_{j=1}^{17} \mu_j \delta_{x_j} = \mu$

• $\sum_{k=1}^5 \xi_k \delta_{z_k} = \xi$

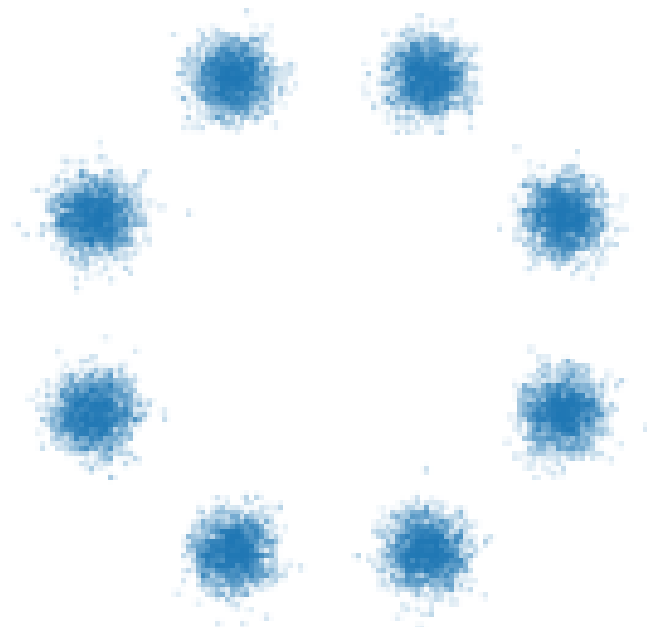
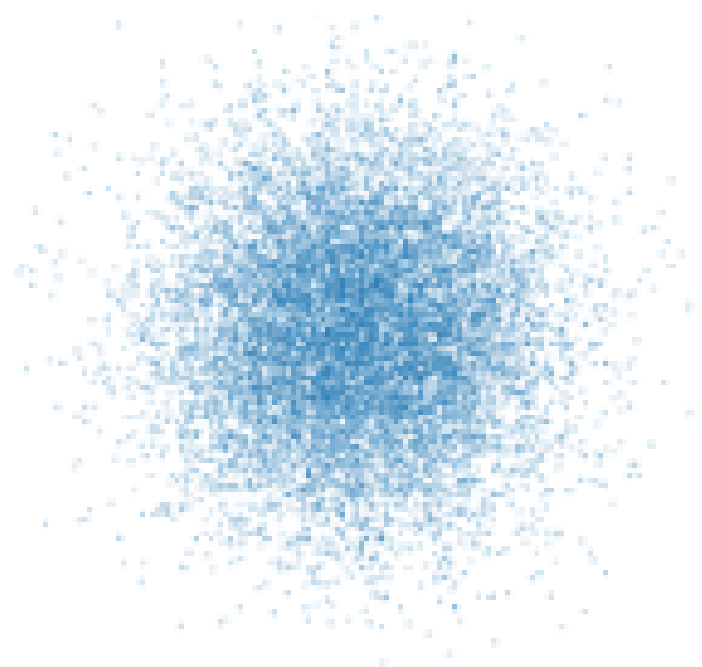


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2	9	3	4	3	9	8	7	2	5
1	5	9	8	3	6	5	7	2	3
9	3	1	9	1	5	8	0	8	4
5	6	2	6	8	5	8	8	9	9
3	7	7	0	9	4	8	5	4	3
7	9	6	4	7	0	6	9	2	3

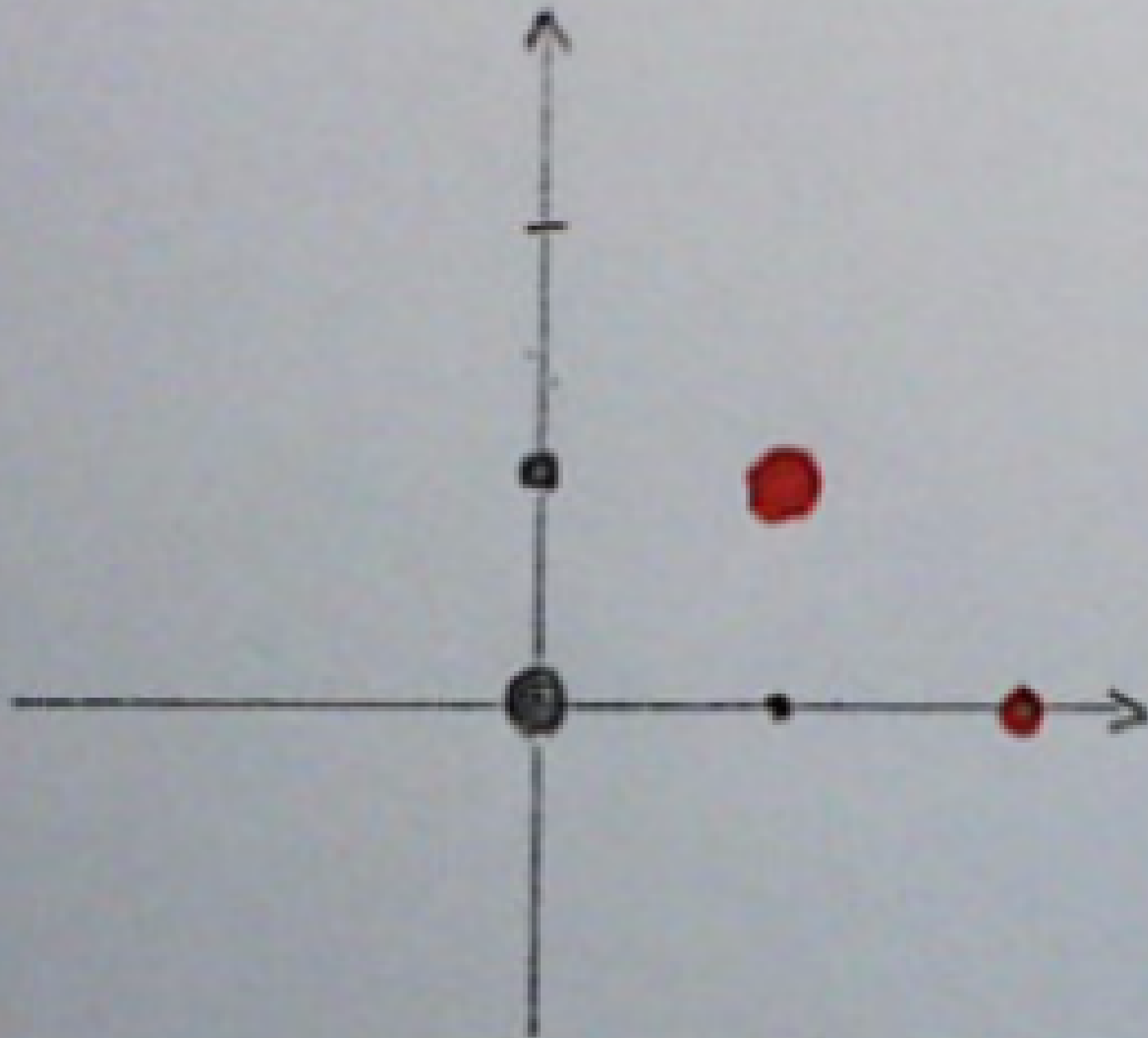




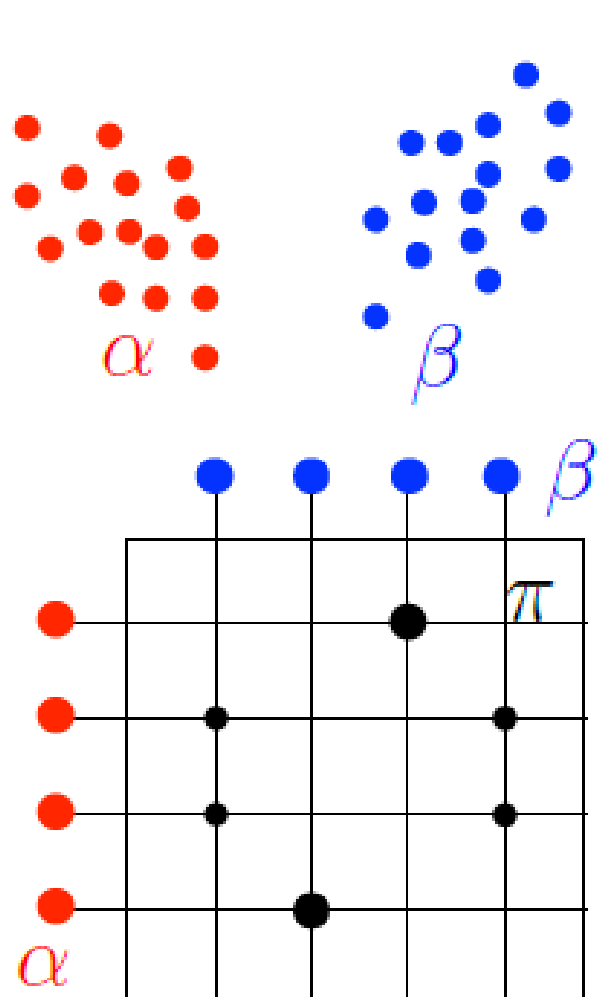




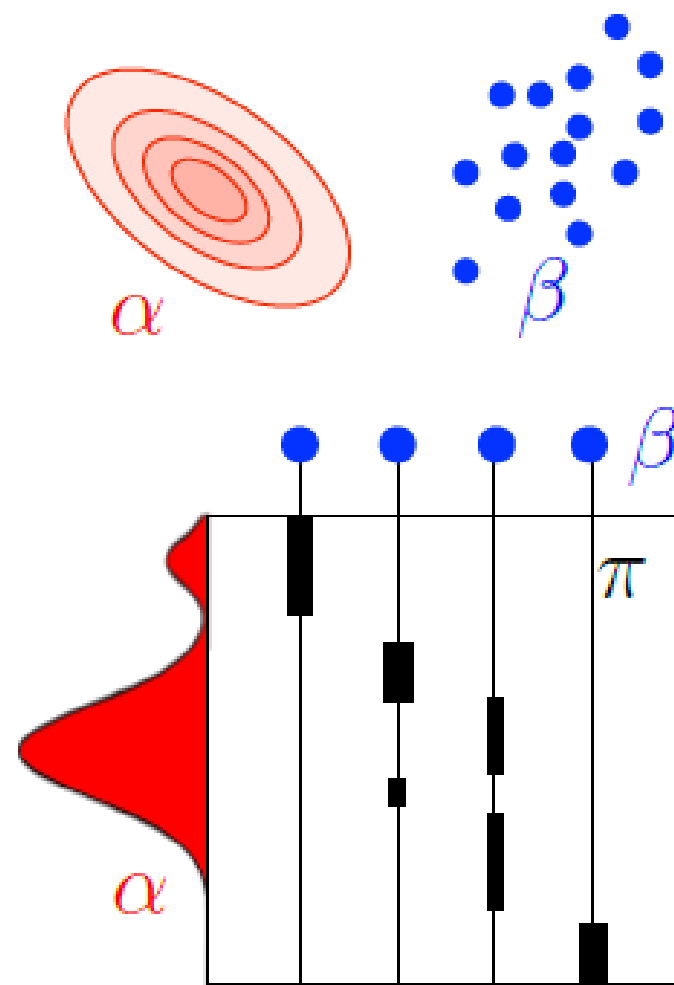




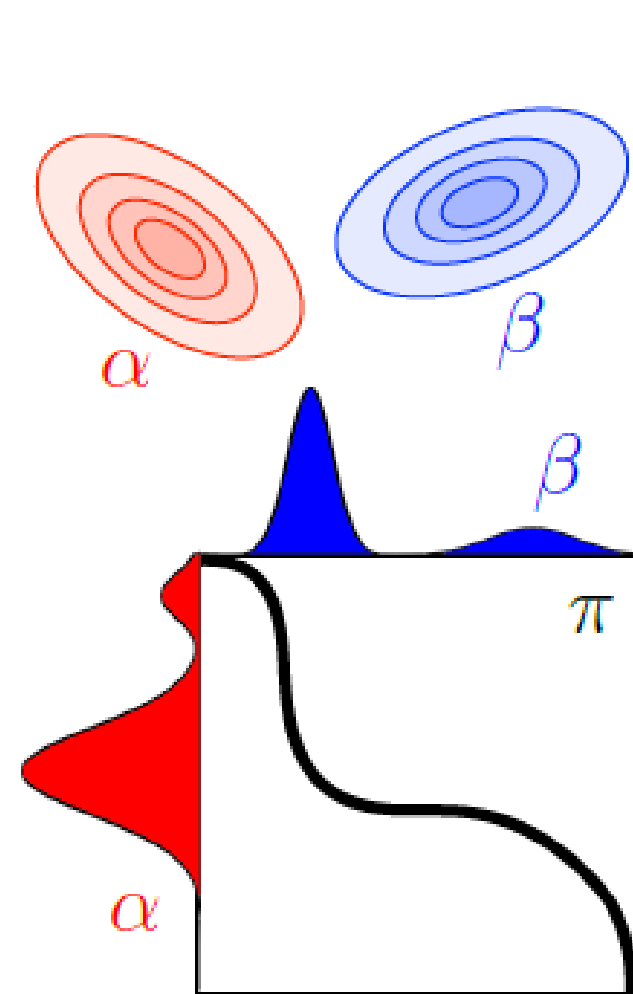




Discrete

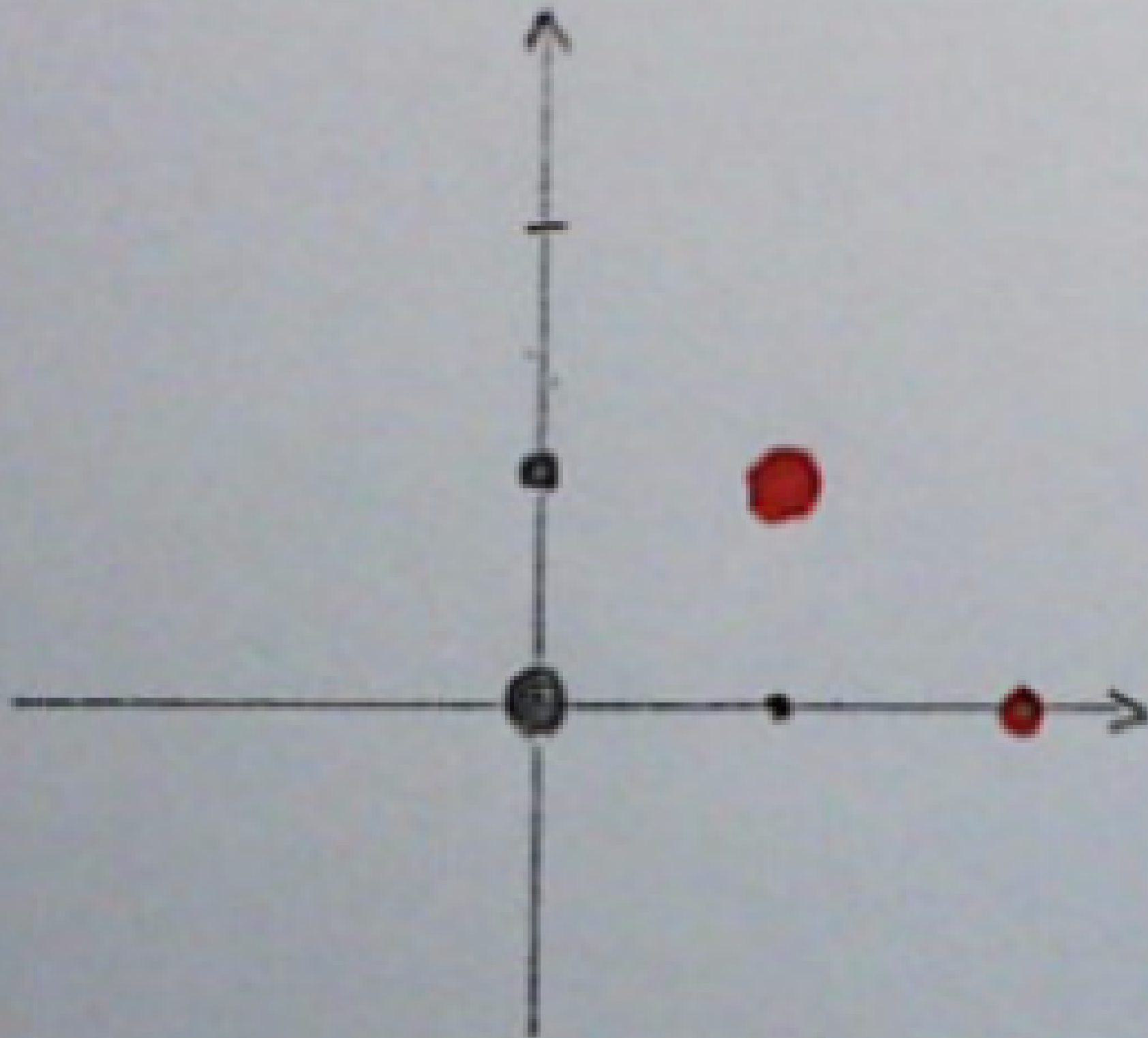


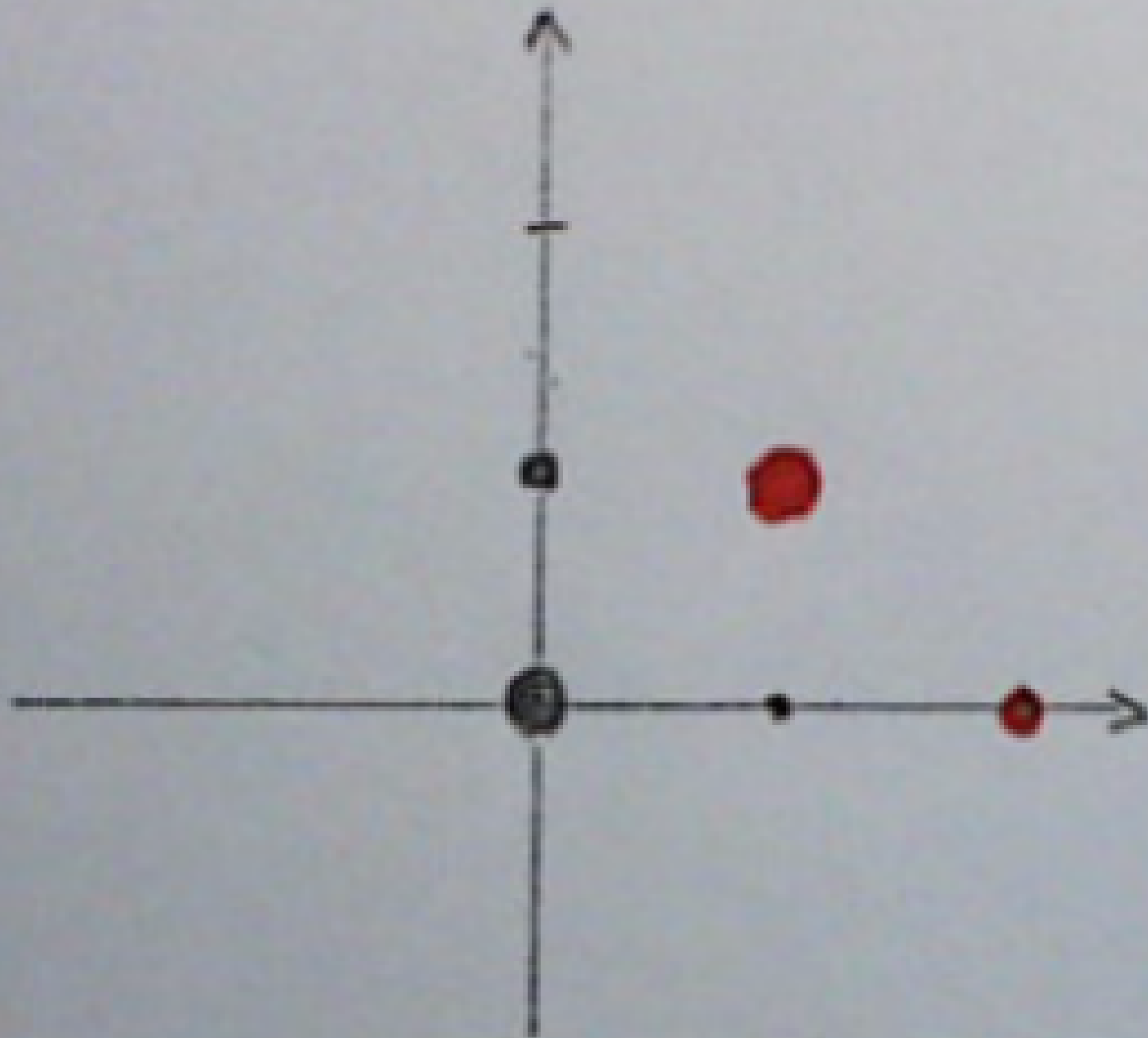
Semidiscrete

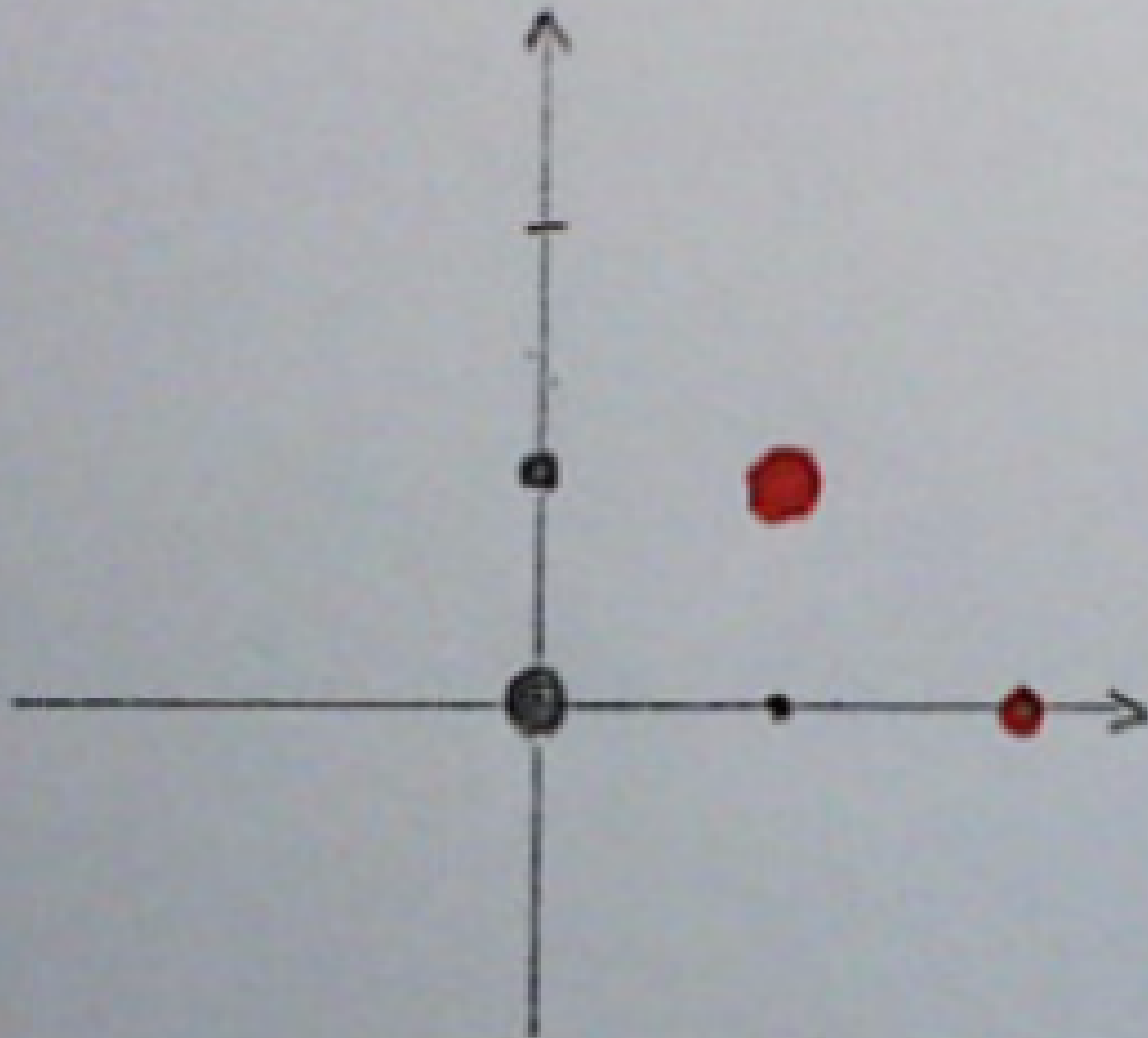


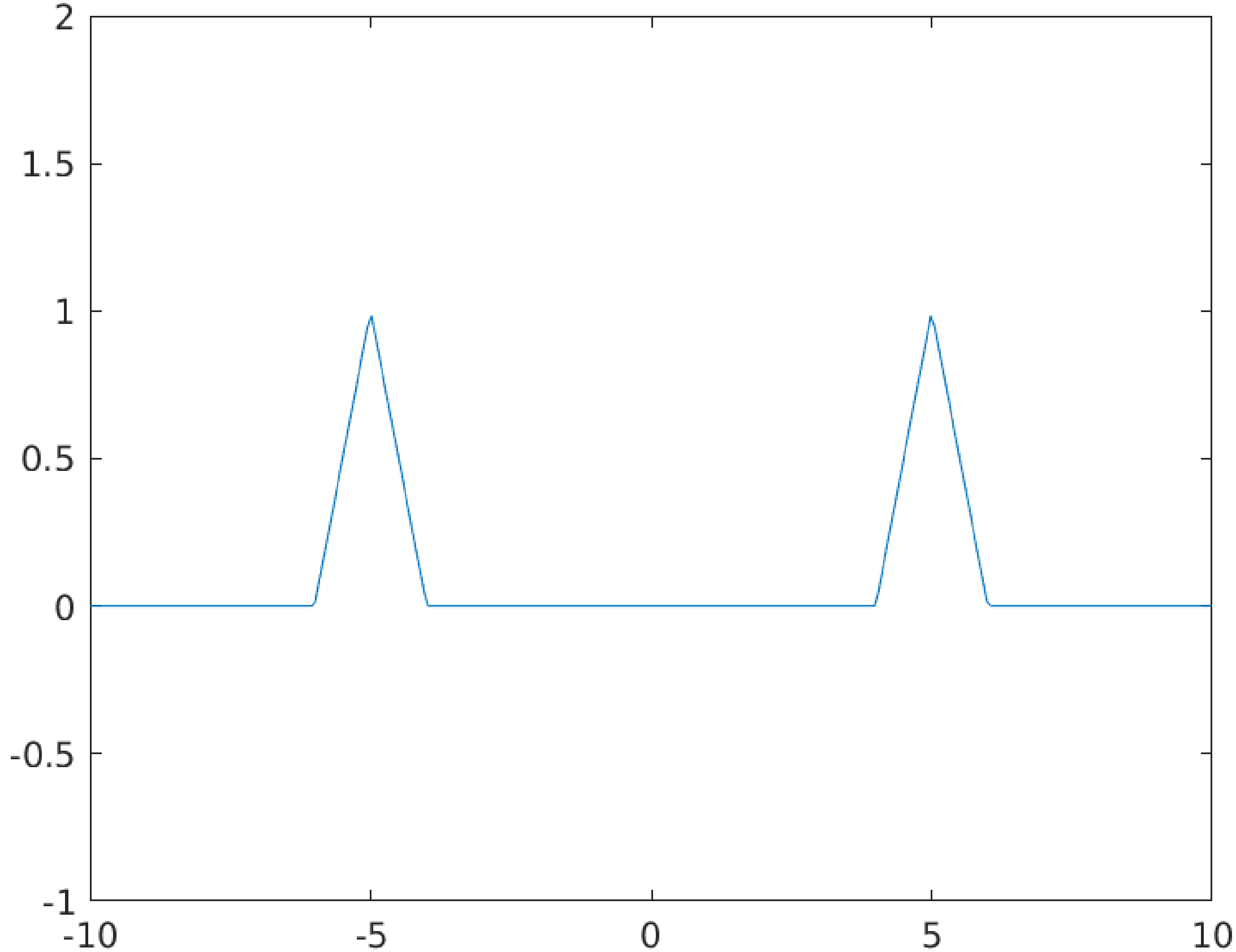
Continuous

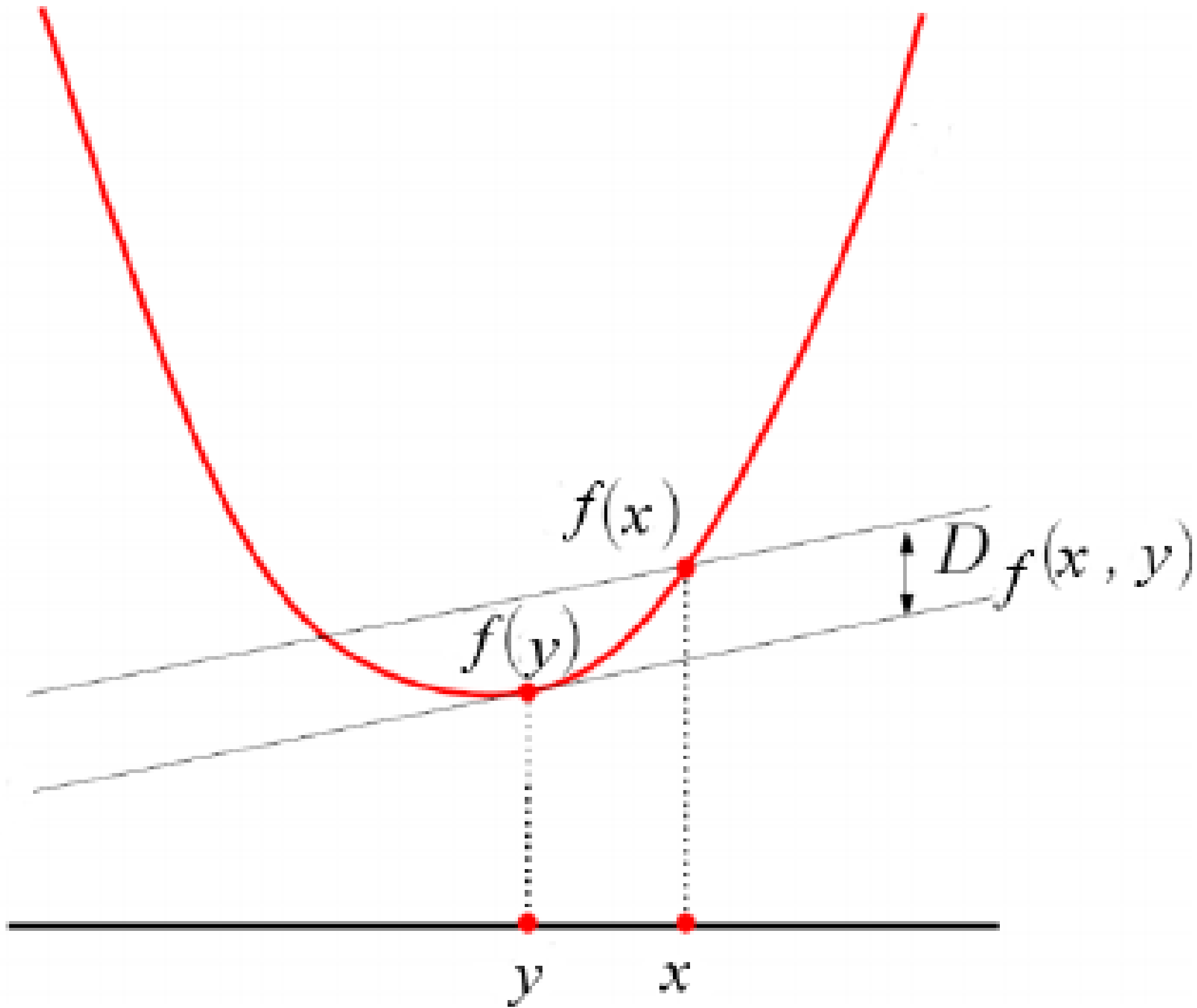












Function Name	$f(x)$	$\text{dom } f$	$D_f(x, y)$
Squared Norm	$\frac{1}{2}x^2$	$(-\infty, \infty)$	$\frac{1}{2}(x - y)^2$
Shannon Entropy	$x \log x$	$[0, \infty)$	$x \log \frac{x}{y} - x + y$
Bit Entropy	$x \log x + (1 - x) \log(1 - x)$	$[0, 1]$	$x \log \frac{x}{y} + (1 - x) \log \frac{1-x}{1-y}$
Burg Entropy	$-\log x$	$(0, \infty)$	$\frac{x}{y} - \log \frac{x}{y} - 1$

$$OT_E(p, 0) = \min_{\pi} KL(\pi, K) \quad \text{st} \quad \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} \pi = \begin{pmatrix} \mu \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} \backslash & \backslash & \backslash \\ - & - & - \end{pmatrix}$$

GENERAL TASK: $A^T 1 = 1$, $A \geq 0$ stochastic matrix, $b \geq 0$

(1) $\min_x KL(x, y) \quad \text{st} \quad Ax = b$

$P_C^{KL}(y)$, $C := \{x : Ax = b\}$ KL projection onto affine set

(2) $\min_{x \geq 0} KL(Ax, b)$

derivation of alg: $\nabla KL(Ax, b) = A^T \log \frac{Ax}{b}$

$$0 = A^T \log \frac{Ax}{b}$$

$$1 = e^{-\tau A^T \log \frac{Ax}{b}}$$

$$x = x_0 e^{-\tau A^T \log \frac{Ax}{b}} \quad \text{fixed point eq.}$$

$$\begin{matrix} x^{(0)} \\ x^{(t+1)} = x^{(t)} \circ e^{-\tau A^T \log \frac{Ax^{(t)}}{b}} \end{matrix} \quad \begin{matrix} \text{Picard it.} \\ \text{SMART} \\ (\text{Byrne 1996, ...}) \end{matrix}$$

relation to (1)

$$x^{(t)} \xrightarrow{t} x^* \in C_{\geq 0} := \{x \geq 0 : Ax = b\} \neq \emptyset$$

$$x^* = \argmin_{x \in C_{\geq 0}} KL(x, x^{(0)})$$

$\uparrow$
y

(3) $\min f(x)$

$x \in C \leftarrow \text{convex, closed}$

- gradient descent reproj. (2) not Lipschitz cont. gradient of f

$$x^{(t+1)} = \argmin \frac{1}{\tau} KL(x, x^{(t)}) + f(x) \quad (\text{BPP})$$

$$\odot \quad 0 = \log x - \log(x^{(t)}) + \tau \nabla f(x)$$

$\downarrow$
 $D_h(x, x^{(t)})$
(more general)

linearization of f :

$$x^{(t+1)} = \argmin_{x \in C} \frac{1}{\tau} KL(x, x^{(t)}) + f(x^{(t)}) + \langle \nabla f(x^{(t)}), x - x^{(t)} \rangle$$

$$= \argmin_{x \in C} KL(x, x^{(t)}) + \tau \langle \nabla f(x^{(t)}), x \rangle$$

Mirror descent alg (MDA)

(Nesterov / Yudin 1993, Beck / Teboulle 2003)

$$x^{(t+1)} = \argmin_{x \in C} D_h(x, x^{(t)}) + \tau \langle \nabla f(x^{(t)}), x \rangle \quad \text{or equivalent}$$

$$= \argmin_{x \in C} h(x) + \langle \tau \nabla f(x^{(t)}) - \nabla h(x^{(t)}), x \rangle$$

$D_h = KL$; $x^{(t+1)} = x^{(t)} \circ e^{-\tau \nabla f(x^{(t)})}$

$f(x) = KL(Ax, b)$; $x^{(t+1)} = x^{(t)} \circ e^{-\tau A^T \log \frac{Ax^{(t)}}{b}}$

Convergence: - general MDA $O(\frac{1}{t^2})$

- accelerated BPP: $O(\frac{1}{t^2})$ $\forall t \in [1, 2]$ for special h
(singular scaling prop. of D_h)

• SMART for OT_E : $A = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix}$ scale by $\frac{1}{2}$ to get $A^T \mathbf{1} = \mathbf{1}$

$$\pi^{(t+1)} = \text{diag} \left(\frac{\nu}{(\pi^{(t)})^T \mathbf{1}} \right)^{1/2} \pi^{(t)} \text{diag} \left(\frac{\mu}{\pi^{(t)} \mathbf{1}} \right)^{1/2}$$

- This alg. is $\sim 1/4$ slower than Sinkhorn alg.

- it works also with $\alpha < 1$ (not $\alpha = 1$!) instead of $\alpha = \frac{1}{2}$;

then it is $\sim 1/2$ slower than Sinkhorn alg. Question 1: Why?

GENERALIZATION

$$C = C_1 \cap C_2 \cap \dots \cap C_N, \quad C_k := \{x : A_k x = b_k\}, \quad A_k, b_k \geq 0, \quad A_k^T \mathbf{1} = \mathbf{1}$$

$$(4) \quad \arg \min_{x \in C} KL(x, y)$$

Iterative information projection:
(Csiszar 1989)

$$\begin{aligned} x^{(0)} &= y \\ x^{(t+1)} &= P_{C_t}^{KL}(x^{(t)}) \end{aligned}$$

see (1) and (2)

$$C_{t+2N} = C_t \text{ (periodic projection)}$$

- convergence for affine sets; not for general convex sets (Dijkstra alg. for convex sets...)

Generalized iterative scaling (GIS):

$$\text{BI-SHART: } \begin{aligned} x^{(0)} &= y \\ x^{(t+1)} &= x^{(t)} \circ e^{-\tau A_t^T \log \frac{A_t x^{(t)}}{b_t}} \end{aligned}$$

one SMART step

• BI-SHART for OT_E : $\pi^{(t+1)} = \pi^{(t)} \circ \text{diag} \frac{\mu}{\pi^{(t)} \mathbf{1}}$

= Sinkhorn alg.

$$\pi^{(t+2)} = \text{diag} \frac{\nu}{(\pi^{(t+1)})^T \mathbf{1}} \pi^{(t+1)}$$

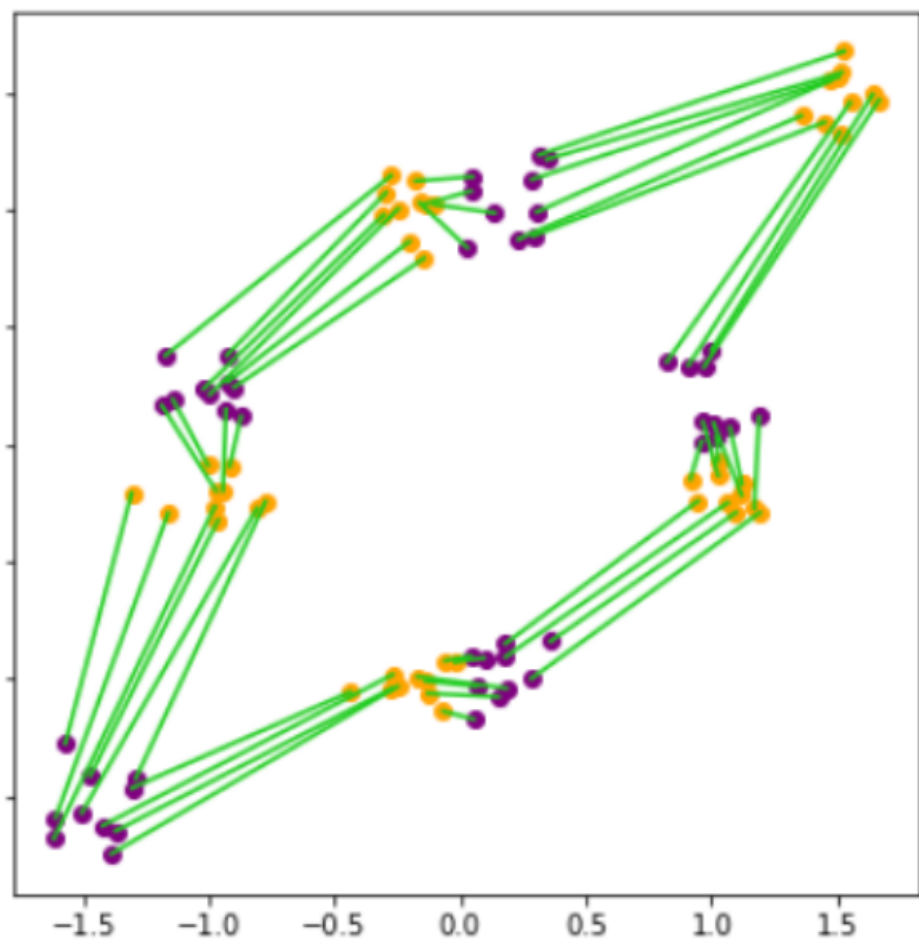
Question 2: BI-SHART for unbalanced OT: $(1-\gamma) KL(x, y) + \gamma KL(Ax, b)$

does not converge in the obvious form

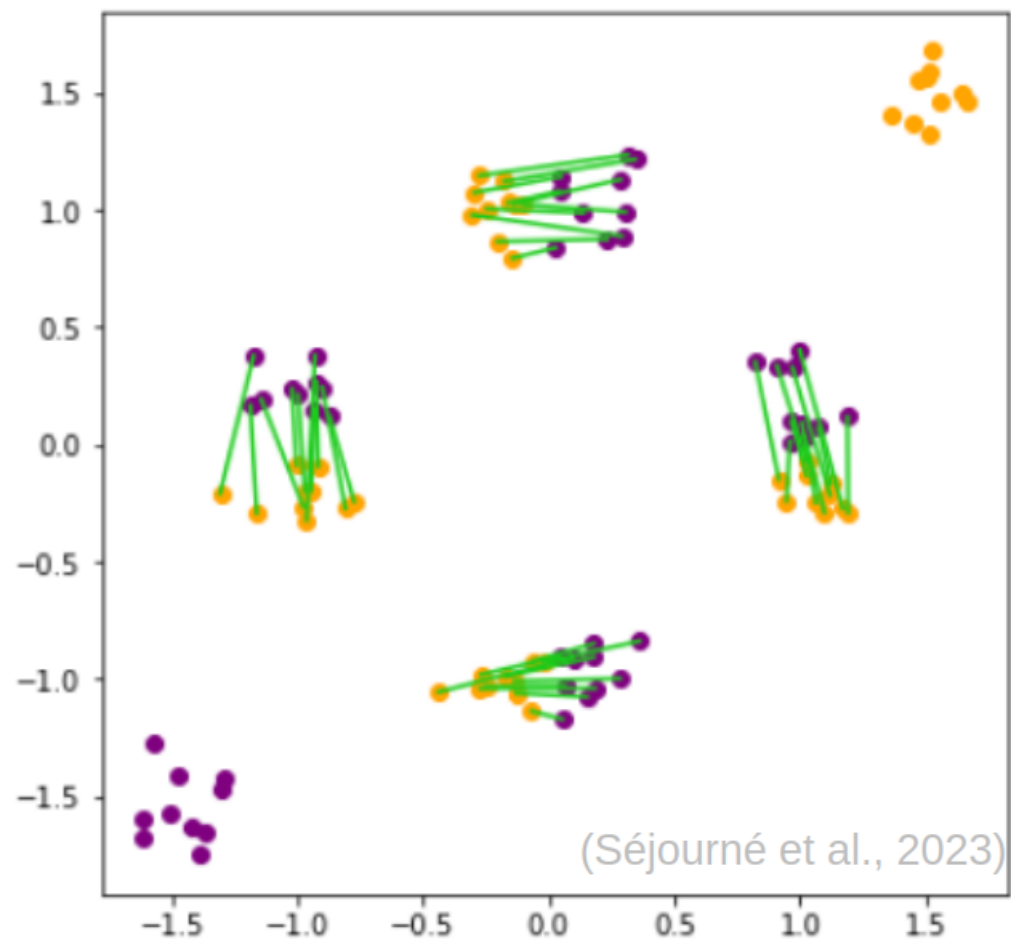
Alternatives?

BI-SHART in case $C = \emptyset$

(old problem, eg. of Byrne)

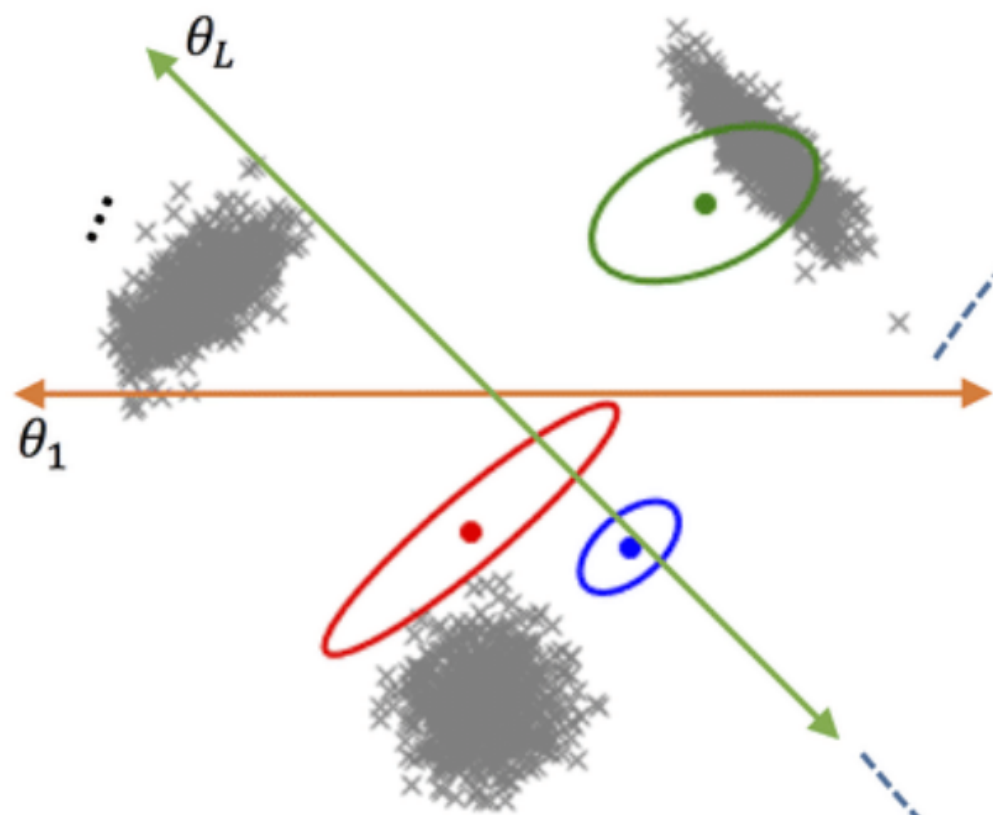


(a) OT matching

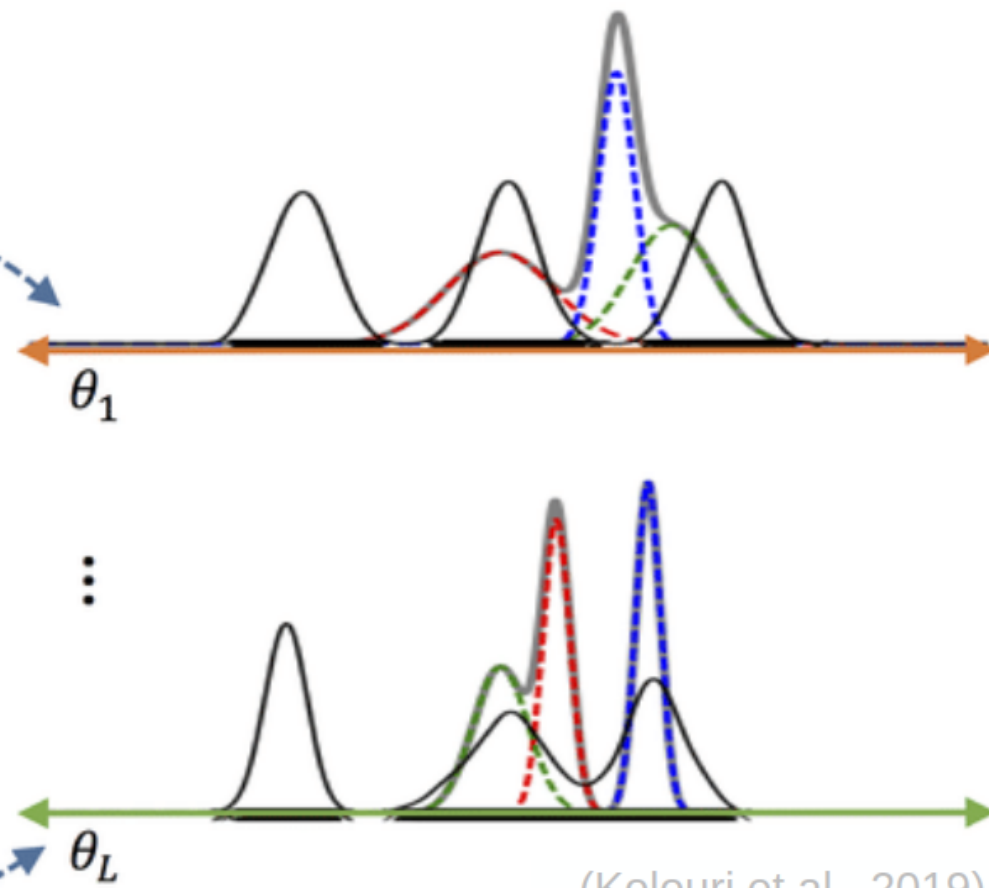


(b) Unbalanced OT matching

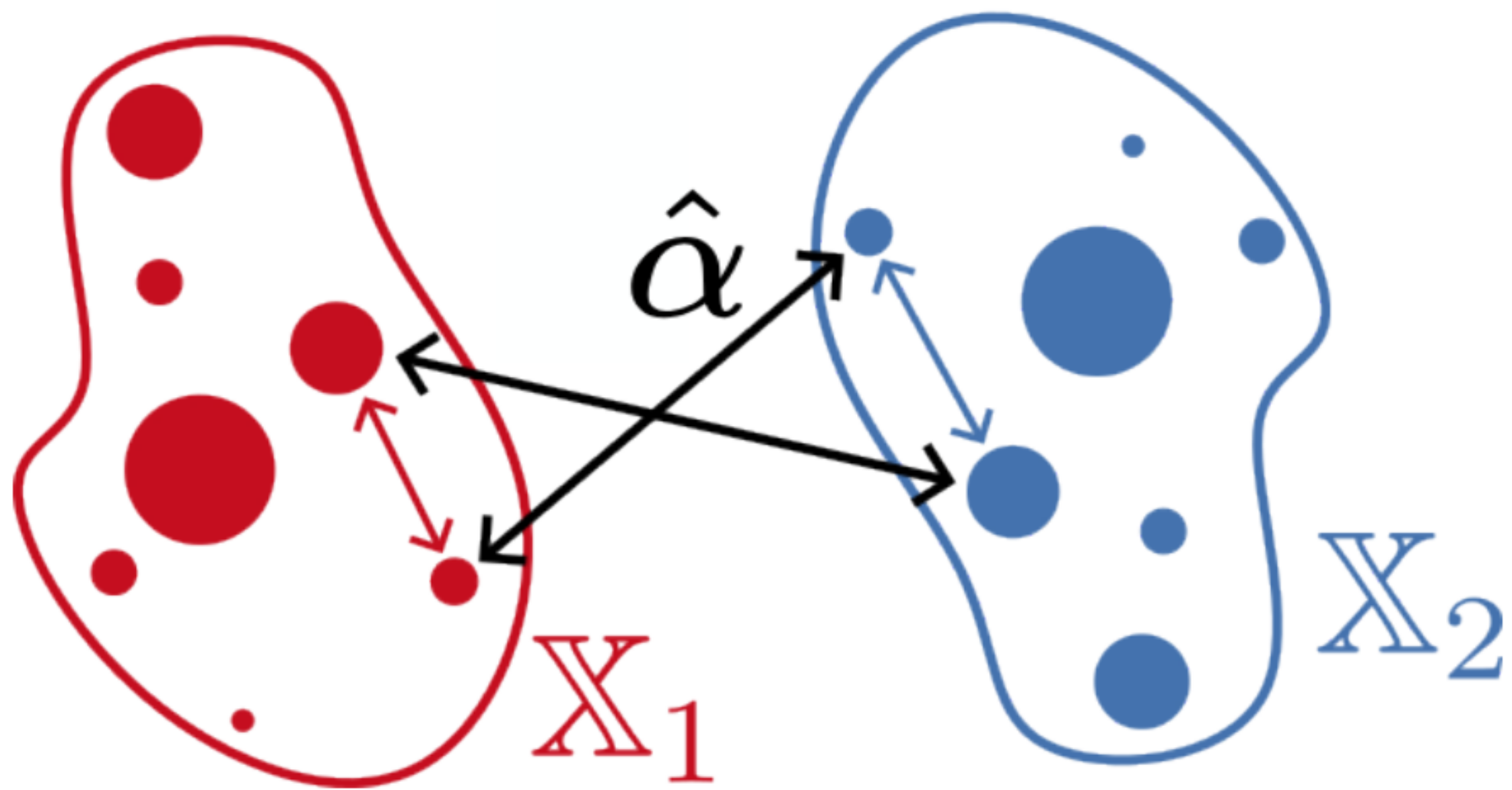
Data space



Projection space



(Kolouri et al., 2019)



1000 x 1000



(Beier & Beinet, 2025)

